

WHAT NOISY FEEDBACK COSTS BAYESIAN OPTIMIZATION, AND WHERE THE COST SITS

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ABSTRACT

Bayesian optimization driven by human ratings is often evaluated against regression oracles fitted to archival data, which mixes the cost of noisy feedback with the oracle’s error. We measure it without an oracle, pairing noisy runs under four error processes on twenty analytic landscapes with identically seeded clean twins. At a rating error of one landscape standard deviation from the first rating, the deployed design loses 29.7% [26.6, 32.9] of the achievable improvement, and 42% [34, 50] of that loss is selection, the gap to the best design the run visited. Tree-ensemble oracles fitted to synthetic ratings of these landscapes report about a quarter of the loss. No search-side change we tested reduced it. A final comparative sitting of two designs, sized on other seeds, recovers about a tenth of it on held-out seeds. If the rater noise of three human studies transfers to this scale, the loss is 22–32%.

1 INTRODUCTION

Bayesian optimization (BO) is the standard tool for optimizing an expensive black-box function from few evaluations (Jones et al., 1998; Shahriari et al., 2016; Frazier, 2018). It is also used for objectives that only a person can evaluate: interface parameters judged for trust or aesthetics, generated content ranked for quality, robot behaviours rated for comfort. This is human-in-the-loop (HITL) optimization (Brochu et al., 2010; González et al., 2017; Koyama et al., 2017; Khajah et al., 2016; Dudley et al., 2019; Koyama et al., 2020; Kadner et al., 2021; Chan et al., 2022; Liao et al., 2023). The optimizer assumes each observation is informative about the objective, but a person is not a sensor. People rate inconsistently, drift, are generous or harsh, make correlated errors, and sometimes apply the wrong setting altogether. How much that costs is asked routinely, but has so far been answered with a design that cannot isolate and answer this question.

A common approach fits a model to archival human responses and treats it as the objective (Siivola et al., 2021; Astudillo et al., 2023; Letham et al., 2022), then injects synthetic error into its responses and measures the degradation (Astudillo et al., 2023). Studies of learning from human preferences use the same design, with a reward model fitted to human labels as the gold standard (Gao et al., 2023). The fitted oracle is itself wrong. In the three archival studies with open data that we draw on (Colley et al., 2025; Jansen et al., 2025; Susak et al., 2026), its cross-validated R^2 is 0.55 at best (ehmi), 0.23 for that study when whole participants are held out, and negative for the other two on the pipeline’s own folds (Appendix B.4). The measured degradation therefore mixes the injected error with the oracle’s misspecification, and regret is measured against a random-search estimate of the optimum that the optimizer can legitimately beat. The experiment also runs on a single design space, so it cannot say whether its acquisition ranking holds elsewhere.

In this work, we replace the fitted oracle with analytic objectives. On an analytic landscape with a published optimum, the objective is exact, regret is measured from the true optimum, and the geometry of the landscape can be measured before the optimization is run. Twenty such landscapes turn “which acquisition is robust here” into a distribution over landscapes, which we report one landscape at a time. Six of them, human-performance laws fixed as a smooth, low-modal subset before any result was computed, stand in for the objectives HITL studies optimize.

Contributions.

1. **A protocol with no fitted oracle, and what error costs HITL optimization.** Twenty analytic landscapes with verified optima, four error processes at two onsets, identically seeded clean twins, and a model-free control. On the same landscapes, a tree-ensemble oracle fitted to synthetic archival ratings, the common evaluation design, reports about a quarter of the deployed loss at 1σ from the first rating. That loss is 29.7% of the achievable improvement, between 18% and 47% on every landscape, and 22–32% at the rater noise of three human studies if their unit transfers (Section 4).
2. **Where the cost sits.** Deployed regret splits into search loss and selection loss (Section 5). Under the standard ship rule, at 1σ selection is 42% of the cost of error, rising to 83% for a large error arriving late, and it is zero under exact observation. Most search-side changes we ran move both terms, and none leads to a gain on the deployed design.
3. **What helps, by how much, and what does not.** We test three procedures that act on the final decision, a comparative sitting, a cautious ship rule and confirmation before a claim of improvement. Every choice among variants is re-made on training seeds and scored on held-out ones. At 1σ from the first rating, a sitting of two designs gains 0.030 [0.012, 0.050] of the achievable improvement on held-out seeds, about a tenth of the loss, and survives Holm’s correction over the nine sitting sizes in that cell. Sitings of sixteen trials or more lose there, and pooled over magnitudes neither the sitting nor the cautious rule survives a family-wise correction (Section 7). Refitting the surrogate on the ratings’ ranks recovers 51% of the cost of gross faults (49% on held-out seeds; Appendices E.4 and E.6).

2 RELATED WORK

Bayesian optimization under noise. Noise has been part of the formulation since Jones et al. (1998) and is routine in practice (Snoek et al., 2012); acquisitions built for it include augmented expected improvement (Huang et al., 2006), the knowledge gradient (Frazier et al., 2009) and noisy expected improvement (Letham et al., 2019; Balandat et al., 2020). Picheny et al. (2013b) benchmark infill criteria under noise on analytic test functions, the closest antecedent; we differ in treating the error *process* and its onset as factors and in asking where the cost sits. Heteroscedastic and non-stationary surrogates (Kersting et al., 2007; Cowen-Rivers et al., 2022), outlier-robust likelihoods (Martinez-Cantin et al., 2018; Ament et al., 2024), corruption-tolerant acquisitions (Bogunovic et al., 2020) and time-varying bandits (Bogunovic et al., 2016) each model one of our error processes; we test representatives of the first two as remedies (Appendices E.4 and E.5). The regret of the best design visited and of the one recommended, simple and inference regret, are both standard in noisy-BO benchmarking (Picheny et al., 2013b); we measure their gap under feedback error. Shipping the best observed rating selects on noise, so the shipped design’s rating overstates its value, the optimizer’s curse of decision analysis (Smith & Winkler, 2006). Selection loss is the regret the same selection on noise causes. The same bias inflates a claim that the shipped design improves on the start, which inference on winners corrects (Andrews et al., 2024) and Section 7 tests on fresh ratings. The ship rule that closes the gap when the surrogate is right, the posterior-mean maximiser, is the one the knowledge gradient optimizes for (Frazier et al., 2009); Section 5 scores it. Spending the end of a budget on choosing among visited designs is best-arm identification (Audibert et al., 2010) and ranking-and-selection (Kim & Nelson, 2006), the clean-up step after simulation optimization (Boessel et al., 2003); kriging-based optimizers with such a step have been compared on test functions under noise (Jalali et al., 2017). We measure what the step is worth under rater error, landscape by landscape.

Human-in-the-loop and preference-based optimization. Where the objective is a person’s judgement, the usual response is to change the observation model: pairwise preferences with an ordinal likelihood (Chu & Ghahramani, 2005; González et al., 2017; Astudillo et al., 2023), or queries a crowd can answer as a line search (Koyama et al., 2017). Chan et al. (2022) document the benefits and drawbacks from the user’s side. Reward models fitted to human labels serve as the evaluation oracle in preference learning at scale (Stiennon et al., 2020; Ouyang et al., 2022; Gao et al., 2023). The oracle-isolation experiment of Section 4 tests the same construction with tree-ensemble oracles on low-dimensional objectives. A comparison is immune by construction to a strictly monotone fault shared across a sitting; this removes a noise-free drift, and Appendix E.7 finds it does not help against a saturating scale.

The landscape as the unit of replication. Results on analytic test functions need not transfer (Eggensperger et al., 2021). The landscape is therefore our unit of replication, and where the observational design cannot identify an effect, we manipulate the geometry directly.

3 SETUP

3.1 LANDSCAPES

Twenty analytic landscapes with verified optima, $d = 2$ to 11: thirteen classical test functions, six human-performance laws including the Yerkes–Dodson inverted-U (Yerkes & Dodson, 1908), and moving peaks (Branke, 1999) (Appendix A.3). The six laws are smooth with at most one interior mode, like the preference and performance objectives HITL studies optimize; four are additive across their inputs and four have a corner optimum (Appendix B.6). We fixed them as the smooth subset before computing any result. Each objective is affinely standardised to zero mean and unit variance over its box on 2^{16} scrambled-Sobol points, so a single grid of error magnitudes serves outputs spanning seven orders of magnitude, with the argmax unchanged. opt_z , the number of landscape standard deviations (landscape SDs) by which the optimum exceeds the mean, runs from 0.76 to 56.8.

3.2 ERROR PROCESSES

Let $g(x)$ be the standardised objective, t the trial, t_0 the onset and σ_e the swept magnitude, in landscape SDs. Observations are exact for $t \leq t_0$. After that, the optimizer sees $\tilde{g}_t$:

$$\text{GAUSSIAN} \quad \tilde{g}_t = g(x_t) + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (1)$$

$$\text{BIAS} \quad \tilde{g}_t = g(x_t) + \sigma_e + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (2)$$

$$\text{DRIFT} \quad \tilde{g}_t = g(x_t) + \sigma_e \frac{t-t_0}{T-t_0} + \epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, \sigma_e^2) \quad (3)$$

$$\text{AR}(1) \quad \tilde{g}_t = g(x_t) + e_t, \quad e_t = \rho e_{t-1} + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma_e^2(1 - \rho^2)) \quad (4)$$

with $\rho = 0.8$ and e_{t_0+1} stationary. BIAS is a generous rater, DRIFT fatigue over a session, AR(1) a rater whose consecutive judgements are correlated. These design choices are checked against three archival studies (Appendix B.4), which show rater offsets and session drift of about this size (though the drift differs between raters). They show little positive serial correlation once each session’s own trend is removed, so that AR(1) is a stress test. They also show heavier tails and more saturation than the main arms have, which the gross-fault and capped-scale arms model. We sweep $\sigma_e \in \{0.05, 0.25, 1, 5\}$, the last a stress level, and $t_0 \in \{0, 20\}$ over $T = 50$ trials with 5 initial random samples, twelve acquisitions and ten seeds: 76,800 noisy runs, each paired with one of 2,400 clean twins (the seeds of every arm are in Appendix A.2). Appendix D.4 adds slips, errors in the design the person experiences, such as a slider that overshoots, a noisy-input setting (Oliveira et al., 2019; Fröhlich et al., 2020) with the perturbation unknown.

3.3 OPTIMIZER AND ACQUISITIONS

Every arm uses BoTorch’s (v 0.18.2) default single-task Gaussian process (GP) (Balandat et al., 2020): a radial-basis-function (RBF) kernel with dimension-scaled lengthscale priors, inputs normalised to the unit cube, outputs standardised, fitted by marginal likelihood. Ten model-based acquisitions, five strategies in analytic, log (Ament et al., 2023) or Monte-Carlo form: expected improvement (EI, LogEI, qEI), probability of improvement (PI, LogPI, qPI), the upper confidence bound (Srinivas et al., 2010) (UCB, qUCB), noisy expected improvement (Letham et al., 2019) (qNEI) and the posterior-mean maximiser (Greedy). There are also two model-free floors, uniform random and scrambled Sobol. Improvement-based acquisitions take the maximum posterior mean at visited points as their incumbent, the usual choice under noise and not a neutral one (Appendix D.2).

3.4 METRICS

The primary estimand is deployed loss: the regret of the deployed design, the visited design with the best observed rating at the final trial, minus the same in the identically seeded clean twin, divided

Table 1: Fraction of the achievable improvement destroyed by feedback error, averaged over landscapes and the ten model-based acquisitions, by the error magnitude in landscape SDs. The two final-trial blocks are on the same time basis: the deployed minus final search loss is the selection loss, and the last block gives it as a share of the deployed excess, with landscape-bootstrap intervals. The trajectory block is a time average and bounded by the clean run’s post-onset improvement (Section 4); it is not comparable with the final-trial blocks.

error present	0.05σ	0.25σ	1σ	5σ
<i>search loss, post-onset per-trial average</i>				
from trial 1	0.9%	5.3%	14.1%	22.0%
from trial 21	0.3%	1.2%	2.8%	4.9%
<i>search loss, final trial</i>				
from trial 1	1.0%	6.2%	17.3%	28.5%
from trial 21	0.3%	1.6%	4.1%	7.5%
<i>deployed design, final trial</i>				
from trial 1	1.5%	9.1%	29.7%	71.1%
from trial 21	0.8%	4.1%	13.6%	44.4%
<i>of that deployed excess, the share that is selection loss</i>				
from trial 1	31% [20, 44]	32% [23, 41]	42% [34, 50]	60% [52, 68]
from trial 21	55% [38, 79]	60% [49, 73]	70% [59, 79]	83% [75, 90]

by the landscape’s achievable improvement opt_z . The tables call it the deployed excess. A *ship rule* is how the deployed design is picked from the visited ones. The best observed rating is the standard rule; Appendix C.1 scores four others, among them the posterior-mean maximiser that noise-aware practice recommends; Section 5 reports what they recover. **Search loss** is the regret of the best design visited, $y^* - \max_{t' \leq t} f(x_{t'})$; at the final trial, deployed minus search loss is the selection loss of Section 5. Its post-onset per-trial average, the *trajectory loss*, has lower variance and drives the acquisition ranking and several appendices, but it is bounded by the clean run’s improvement after the onset (Section 4). A final value is compared only with a final value. Paired runs share their candidate-proposal stream, and y^* is a verified supremum. Since a landscape SD is not a fully comparable unit across landscapes (Appendix B.2), every aggregate is formed from per-landscape values and every headline is also shown per landscape. Intervals are 95% percentile landscape bootstraps; at twenty clusters they cover 92–93%, and a t interval that covers about 95% is up to two points wider (Appendix A.2).

3.5 CONTROLS

Appendix A.1 details three controls: the model-free floors never read an observation, and their excess search loss is identically zero over all 12,800 comparisons; no landscape is ceiling-limited by its clean run; and BIAS from the first rating matches GAUSSIAN to within 0.012 at every magnitude, as a GP that standardises its targets requires.

4 HOW MUCH DOES FEEDBACK ERROR COST?

Table 1 and Figure 1 give the cost at every magnitude and onset. Error whose standard deviation equals the landscape’s own, present from the first rating, costs the deployed design 29.7% [26.6, 32.9] of the achievable improvement. The search loss is 17.3%, so the deployed loss is $1.7\times$ the search loss, and $2.5\times$ at 5σ , where the deployed loss reaches 71.1%. Figure 2 shows the same cell landscape by landscape: the deployed loss is between 18% (Rosenbrock) and 47% (Hartmann-6) on every landscape (median 29%), and every interval excludes zero. The six human-performance laws lose 32.7% [29, 37] against 27.8% [24, 32] for the thirteen classical functions, so the cost is not a product of multimodality (Appendix B.6).

The late onset looks cheap on the trajectory for a mechanical reason. There, 1σ error from trial 21 costs 2.8% against 14.1% from the first rating, an onset ratio (cost from the first rating over cost

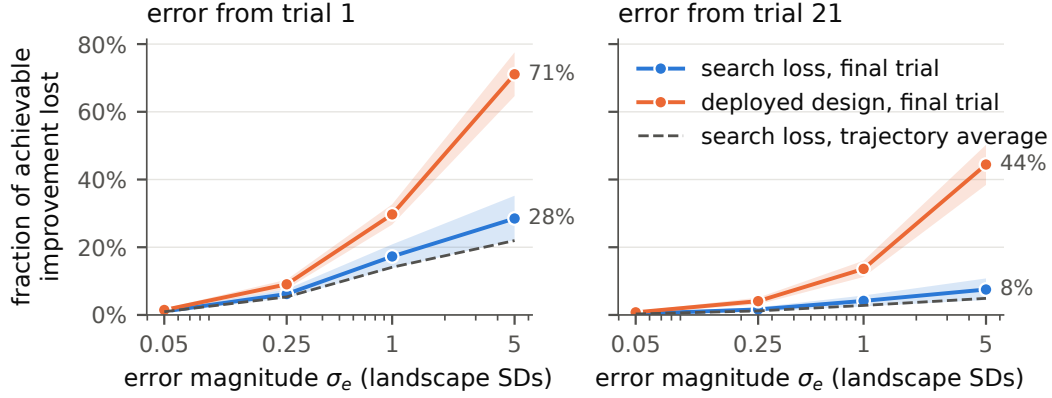


Figure 1: Deployed loss and search loss at the same final trial against the error magnitude; their gap is the selection loss. The dashed line is the trajectory average. Twenty landscapes, ten model-based acquisitions, 95% landscape-bootstrap bands.

from trial 21) of $5.0\times$. But a noisy run and its clean twin share every design until the error arrives, so the post-onset excess can never exceed the improvement the clean run still makes after that point. In 27% of late-onset runs at 1σ it sits on that bound throughout (Appendix B.3). The bound is $5.3\times$ smaller at the late onset, and divided by it, error from trial 21 destroys 43% [34, 45] of the improvement still available and error from the first rating 41% [36, 45], a ratio of 0.95 [0.78, 1.20]. The bound also produces the growth of the onset effect with the budget, from $2.3\times$ at $T = 25$ to $11.7\times$ at $T = 100$ (Appendix B.8). The deployed design is not bounded this way: there error from trial 21 still costs 13.6% at 1σ , about half the cost from the first rating.

Of the five factors, the magnitude matters most. Over the 6,400 cell means of the model-based sweep, the magnitude accounts for 24% of the variance of the trajectory loss, the onset for 17%, the landscape for 10%, the acquisition for 4% and the error process for 0.2% (main-effect shares in a balanced design; the rest is interaction). Within the 1σ , first-rating cell the landscape accounts for 49%, the acquisition for 22% and the error process for 2%. The error process’s small share is partly fixed by construction (BIAS from the first rating is GAUSSIAN plus an offset that the standardised GP cannot see), so it describes these parameterisations rather than kinds of human error.

Is 1σ a lot? Within-rater noise in three published human studies (Colley et al., 2025; Jansen et al., 2025; Susak et al., 2026) is estimated as a nearest-neighbour nugget (Cressie, 1993) over ratings of close designs by the same participant and expressed in σ_f , the spread of each study’s fitted objective. It is 0.74 [0.48, 0.97], 1.11 [0.81, 1.40] and 0.60 [0.00, 0.90] σ_f , with participant-bootstrap intervals (Table 5); σ_f plays the part the landscape SD plays in the sweep. As logged, two defects in the archival composites inflated the second and third estimates to 3.35 and 1.02 (Table 5, corrected in Appendix B.4); once fixed, neither study’s oracle predicts held-out ratings, so their σ_f is fragile (another oracle family puts the second estimate at 0.67). Participants are noisy from their first rating, so on the rows of Table 1 for error from the first rating these studies would lose 22–32% on the deployed design, interpolating log-linearly and taking σ_f as the landscape SD. The nugget can understate the noise, and the within-session residual bounds it at $1.2\text{--}2.4\sigma_f$, where the loss would reach up to about half. The anchor fixes the range of interest.

Does a fitted oracle understate it? The fitted-oracle design is the common alternative, and its own companion arm on the three archival datasets reports a third to a half of the cost measured here from the first rating (Appendix B.5). But three human objectives set against twenty analytic ones are different landscapes. We therefore hold the landscape fixed and change only the oracle. For each landscape, we build a synthetic archival dataset as an archival study would produce it (37 participants, each rating the first 20 designs of a clean BO run once, with noise at $1\sigma_f$). We fit an oracle to it using the pipeline’s own model selection, whose seven candidate families are all tree ensembles (it picks gradient boosting on 15 landscapes, extra trees on 3, CatBoost on 2). We then rerun the experiment against that oracle, with the error scaled by its own σ_f and regret

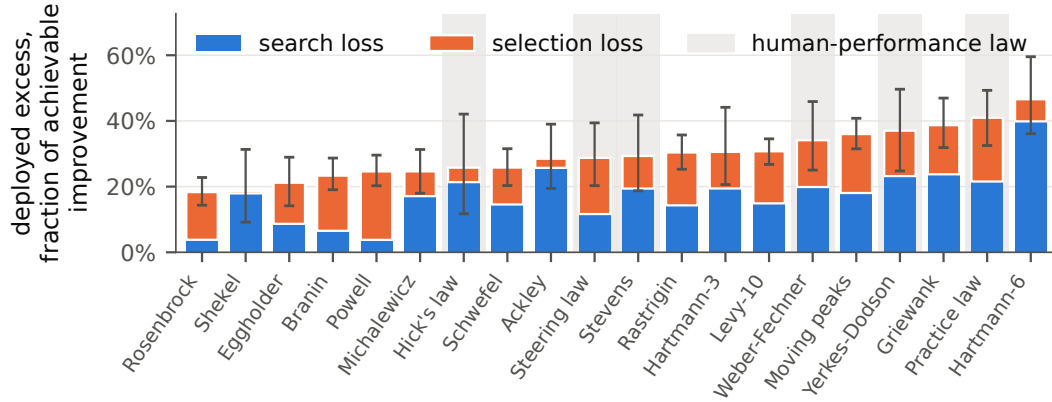


Figure 2: The headline cell per landscape: the deployed design’s excess over its clean twin at 1σ from the first rating, split into search and selection loss (Equation 5), pooled over the four error processes and ten acquisitions. Bars are the deployed excess with 95% intervals over acquisitions and seeds, since a single landscape’s bar cannot resample landscapes. Shaded columns, labelled in the legend, mark the six human-performance laws, the smooth subset fixed in advance.

measured relative to the best value any clean run reached. On the deployed design at 1σ it reports 0.25 [0.14, 0.78] of what the exact objective on the same landscape shows, less on 15 of the 20 landscapes (Wilcoxon $p = 0.014$). On the trajectory, it reports 17.7% of the gap between the optimum and a model-free floor against 57.9%, a ratio of 0.31 [0.20, 0.52], less on 18 of 20 ($p = 0.0003$). The ratios at 0.25σ and 5σ are similar or smaller. The oracles are not poor (median correlation 0.84 with the true landscape), and on the deployed design, the better ones understate more (Spearman -0.50). On the trajectory, it does not rank the landscapes: its per-landscape cost correlates 0.11 with the exact one. Nor does it rank the acquisitions as the exact objective does. Under GAUSSIAN error from the first rating, the exact arm puts qNEI and UCB first (mean rank 2.2 of 4) and LogEI last (2.9); the fitted-oracle arm puts LogEI first (2.3) and qNEI last (2.7), and the most robust acquisition of a landscape agrees between the arms in 13 of 60 cells, about the 15 expected by chance (Appendix B.5). Gaussian-process and neural-network oracles are untested.

5 WHERE THE COST IS: SEARCHING OR SELECTING?

Write x^* for the optimum, V for the designs a run visited and d for the one it deploys. Then

$$\underbrace{f(x^*) - f(d)}_{\text{deployed regret}} = \underbrace{f(x^*) - \max_{x \in V} f(x)}_{\text{search loss}} + \underbrace{\max_{x \in V} f(x) - f(d)}_{\text{selection loss}}. \quad (5)$$

Equation 5 is an accounting identity, useful as a pair of counterfactuals. Selection loss is the amount that a ship rule that knew the true value of every visited design would recover without a single new trial, so it bounds the most that any change to how a study chooses can gain. Search loss is what would remain with that perfect choice, so it bounds what a better search can gain on the deployed design. The selection term is zero under exact observation (the best-rated visited design is then the best visited design). That is why it never appears in a noiseless experiment, and why a study that reports its best-rated design is safe only there. As a check, the largest selection loss over the 2,000 clean runs of the ten model-based acquisitions is exactly zero.

A large part of what error costs is selection, and how large depends on the error. Against the identically seeded clean twin, the selection share rises with the magnitude and with the lateness of the onset, from 31% [20, 44] at 0.05σ from the first rating to 83% [75, 90] at 5σ from trial 21 (Table 1). At 1σ from the first rating the deployed loss of 0.297 splits into 0.173 of search loss and 0.124 of selection loss, so selection is 41.8% [34.1, 50.4]; per landscape it runs from 4% (Shekel) to 85% (Powell) (Figure 2). Pooling every cell would give 61.8%, but that pool is dominated by the 5σ cells and we do not use it.

Table 2: Acquisition robustness, ranked over landscapes within each condition and averaged over conditions, on the trajectory and on the deployed design. Excess is the cost of error as a percentage of the achievable improvement; absolute is the noisy run’s trajectory regret. The floor’s deployed excess is the cost of error to the design a model-free floor would ship.

acquisition	search loss, trajectory			deployed design	
	mean rank	excess	absolute	mean rank	excess
qUCB	3.95	4.14%	27.9%	5.12	22.2%
qNEI	4.34	4.89%	25.2%	5.94	23.9%
UCB	4.41	4.73%	27.4%	4.99	21.5%
LogEI	4.61	5.21%	26.0%	4.85	20.8%
qEI	4.73	5.37%	25.5%	5.27	21.1%
EI	4.98	5.42%	26.2%	5.21	21.4%
qPI	6.29	7.90%	31.0%	5.52	21.4%
Greedy	6.56	8.11%	32.5%	5.20	19.5%
LogPI	7.48	9.19%	31.3%	6.20	22.4%
PI	7.66	9.44%	31.7%	6.69	23.6%
model-free floor	–	0.00%	41.0%	–	17.8%

Search-side changes mostly move both terms, and none nets a gain. Six search-side changes (augmented EI, Thompson sampling, a lower-confidence-bound incumbent, minimum-distance proposals, a slip-aware acquisition and redistributed rating effort) ran as their own arms, with the human trials fixed, each scored against the standard process on the same cells and acquisitions (Appendix E.4). None lowers the deployed loss, and, split by Equation 5, they fail: Thompson sampling (Thompson, 1933; Russo et al., 2018) raises search loss by 5.7 and selection loss by 7.7 points relative to the achievable improvement. Minimum-distance proposals lower search loss by 1.9 [0.2, 3.7] and raise selection loss by 5.7, a better search that ships a worse design. A lower-confidence-bound incumbent, the one change that lowers selection loss (2.2 [0.7, 4.0]), raises search loss by 7.6. Redistributing rating effort across trials raises selection loss by 4.6 to 10.4 points, only one interval excluding zero.

A noise-aware ship rule recovers little, except where the surrogate fits. Under GAUSSIAN error at 0.25σ and above, four other ship rules recover from -1.4% (the mean rating) to $+10.1\%$ (the posterior mean less two latent SDs, the cautious rule of Section 7) of the deployed loss. The posterior-mean maximiser recovers $+0.7\%$ $[-4, +6]$, although at $n = 50$ the GP’s noise estimate has largely recovered (0.85 of the injected SD). On the six laws that rule recovers $+11.5\%$ [6, 16] and beats the best observed rating on all six, so even where the surrogate fits it leaves about nine tenths of the deployed loss in place; on the thirteen classical functions it recovers -3.3% . The GP’s fit shows: on clean runs of the ten rugged classical functions, it reads their roughness as noise (fitted noise SD 0.30 where the truth is zero, against 0.03 on the laws), and its R^2 at fresh points is -0.24 against 0.67. Across landscapes its leave-one-out fit predicts the rule’s gain at Spearman 0.62. The posterior does not inherit the error in general. It is misspecified where the landscape is rugged, and misled by structured error, losing even on the laws under large late DRIFT (-15.9% ; Appendix B.6).

6 WHICH ACQUISITION FUNCTION?

On the trajectory, the confidence-bound family and noisy expected improvement are the most robust, and the probability-of-improvement family and Greedy are the least (Table 2); the Friedman test (Friedman, 1937) over landscapes reaches $p < 0.05$ in 28 of 32 conditions, and the same 28 after a Benjamini–Hochberg correction over the 32 (Holm keeps 25). Landscapes disagree either way (median Kendall’s W (Kendall & Babington Smith, 1939), 0.247 over arms, 0.233 over families), so a winner found on one landscape largely does not transfer.

On the deployed design, the ranking largely dissolves. Ranked on deployed loss, the mean ranks span 4.9 to 6.7 rather than 3.9 to 7.7, the two rankings correlate at Spearman 0.59, and Greedy, eighth on the trajectory, has the lowest deployed loss. With error from the first rating, the model-

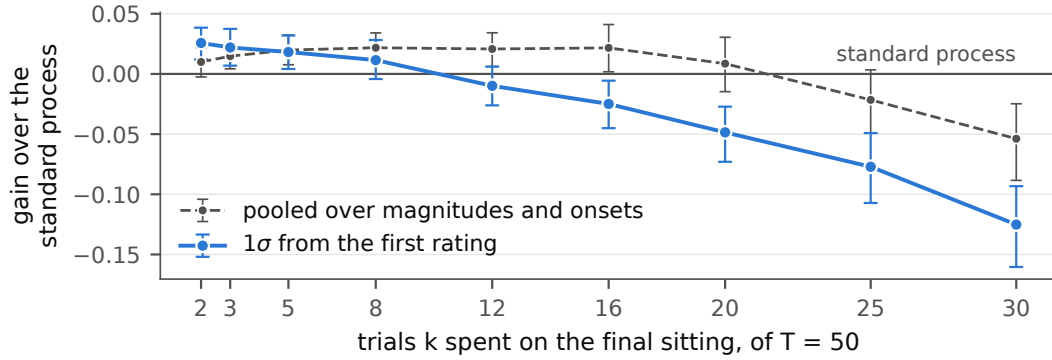


Figure 3: Gain over the standard process from spending the last k of $T = 50$ trials on one comparative sitting among the top- k visited designs, in units of the achievable improvement, with 95% landscape-bootstrap intervals over all ten seeds. Blue, error of 1σ from the first rating. Grey, pooled over the four magnitudes and both onsets. Both pool the four error processes, LogEI and qNEI, and the looks carry the full idiosyncratic SD.

free floors (random and Sobol, averaged) ship a design at 1σ almost as good as PI’s and Greedy’s, and at 5σ a better one than four of the ten acquisitions.

7 BUYING IDENTIFICATION WITH THE TRIALS YOU HAVE

If much of the cost is selection and a better ship rule recovers little, the remaining lever is to spend human trials on identifying rather than searching. Every procedure below changes only the end of a run, so each is replayed exactly from an existing log (Appendix E.1). Each choice we made among variants is re-made on seeds 7–11 and scored on seeds 12–16, with landscape splits as a second check (Appendix E.3, Table 30).

A final comparison. We hold $T = 50$ and spend the last k trials on one sitting. The top k visited designs by the posterior mean less one latent SD are each shown once, and the best-rated is deployed. Error shared across one sitting cancels when designs are compared, and the looks carry the full idiosyncratic noise. The useful size depends on the magnitude and the onset (Figure 3, Table 31). At 1σ from the first rating, the regime of the anchor in Section 4, seeds 7–11 choose $k = 2$, and so do seeds 12–16. For LogEI and qNEI it gains $+0.030$ [$+0.012, +0.050$] of the achievable improvement on seeds 12–16, 12% [5, 18] of the cost of error, on 14 of 20 landscapes and with a positive mean under each of the four processes. Within that cell it survives Holm’s correction over the nine values of k ($p = 0.03$), and random landscape splits choose it in 88% of folds. Sitzings of sixteen trials or more lose there, and twelve loses on average. Pooled over magnitudes and onsets the curve is flat at $+0.020$ to $+0.022$ from $k = 5$ to 16, and most of that gain comes from large errors arriving late. Without the 5σ cell from trial 21, the pooled gain of $k = 12$ is 0.000 [$-0.012, +0.012$] (Appendix E.3).

A model-based choice of k does worse than a fixed one. Choosing k per run from the surrogate’s posterior and the run’s rate of improvement (Appendix E.2) gains -0.007 [$-0.022, +0.006$] on the held-out seeds, 0.024 [$0.009, 0.039$] less than the pooled $k = 12$ chosen on seeds 7–11; in 74% of runs it ships the posterior-mean design, trusting the posterior Section 5 found misspecified.

A cautious ship rule, at no extra trial. Shipping the visited design with the highest posterior mean less two latent SDs, a kriging-quantile rule (Picheny et al., 2013a), recovers 10.1% [4.6, 15.9] of the deployed loss under GAUSSIAN error at 0.25σ and above, 10.7% [5, 16] on the held-out seeds, and 13.9% [11, 17] on the six laws (Appendix B.6). It pays a fixed price of 0.015 whether or not error is present, so it should not be used without an estimate of the rater’s noise.

Confirm before claiming. A study also claims that its design beats where it started. The standard process claims an improvement in 98% of runs, and 15.7% of runs make a false claim, 42.7% at 5σ from the first rating (GAUSSIAN error, LogEI, qNEI and UCB). Spending four final trials rating the chosen and the initial design twice each and claiming only if a one-sided paired t -test at $\alpha = 0.05$

prefers the chosen one takes the false-claim rate to 0.5%. Of that 15.2-point reduction the fresh ratings supply 9.1 points (60%) and the test 6.1. The test has a price. Claiming on the fresh mean alone detects 88% of the real improvements, and the test cuts that to 51%, and to 26% at 1σ from the first rating (Appendix E.1).

Pooled over magnitudes, neither the sitting nor the cautious rule survives a family-wise correction. On the held-out seeds, under GAUSSIAN error with LogEI, qNEI and UCB, the pooled sitting over the top five gains on 15 of 20 landscapes and the cautious rule on 16. With a Wilcoxon test across landscapes and Holm’s correction (Holm, 1979) over the 27 procedures of that family, neither survives (adjusted $p = 0.09$ and 0.06), and with Holm over all 41 procedures no fixed k gains. Of the fixes that spend no trials (Appendix E.6), choosing on the ratings’ ranks recovers 51% [36, 63] of the cost when 15% of trials carry a gross fault (49% [34, 63] on held-out seeds) and nothing when none do.

8 DISCUSSION

If the rater noise measured in three human studies transfers to the landscape scale, feedback error costs the deployed design 22–32% of the improvement it could have had, compared to a clean run that ships within 13% of the optimum (a deployed regret of 0.425 at 1σ , less the 0.297 excess). The oracle-isolation result bears on evaluations that treat a tree-ensemble oracle fitted to human ratings as the ground truth, the construction of Section 1. On the same landscapes such an oracle reported about a quarter of the deployed loss the exact objective showed at 1σ . Three consequences follow from this. Report the deployed design, since the trajectory ranks acquisitions differently and inflates the onset effect (the rankings correlate at 0.59, and the onset ratio is $5\times$ against $2.2\times$). Estimate the rater’s noise relative to the objective’s spread before the study. The magnitude decides more than the kind of error does, and with the onset it decides the useful size of a final sitting. A one-shot predictor computed from the objective adds to the magnitude on the trajectory loss without replacing it, and a clean ten-trial pilot approximates that predictor (Appendices C.2 and C.3). And spend the end of the budget on the decision. A confirmation test before any claim of improvement cuts false claims thirtyfold. At moderate noise from the first rating a sitting of two designs recovers about a tenth of the loss, and larger sittings pay when the noise is large or arrives late. Two mechanisms remain open: the growth of the cost with opt_z beyond what a scale-free error allows (Appendix D.3), and the incumbent’s opposite effects on the deployed design and the trajectory (Appendix D.2).

Limitations. The landscapes are analytic and the raters simulated. The loss appears on every landscape and, from the first rating, is as large on the six laws as on the classical functions (32.7% against 27.8% at 1σ , Appendix B.6). Its transfer to people rests on the noise ratio the archival studies measure. They place it at about 0.6 to $1.1\sigma_f$, inside the sweep, and identify the serial structure less well. The anchor passes through a fitted objective’s σ_f , and the cost at a given magnitude also depends on a landscape’s opt_z (Appendix B.2), which the anchor does not carry. An anchor from repeated ratings is future work. The four main processes are additive and unbounded. A scale capped at the landscape’s 0.9 quantile, which they omit, costs 185% more (Appendix E.4), so the headline may understate the loss on scales that saturate. The oracle-isolation arm uses tree-ensemble oracles only. The procedures of Section 7 were chosen from families of variants. Held-out seeds and landscape splits bound the optimism of those choices at 0.006 of the achievable improvement for k and 3.5 points of recovery for the sitting, with the family-wise tests beside them (Appendix E.3). The per-magnitude analysis of the sitting followed the pooled one. In the reported cell its choice of k agrees in both seed directions, and over all eight cells Holm’s adjusted p is 0.23. Of the observation-model remedies the rank rule is scored on held-out seeds, and clipping, relevance pursuit and per-rater offsets ran on five seeds. The sitting model cancels shared error and has no order effects, which a real sitting would counterbalance. The multi-objective arm cannot separate the hypervolume acquisitions, and the preregistered replication is void on its own control rule, so Appendix E.3 carries the out-of-sample evidence.

AI USE STATEMENT

In this work, we used generative AI tools (Claude, Anthropic) for the following tasks with required disclosure: designing and giving feedback on the research methodology and the experiments; imple-

menting methods, including the simulator, the analysis pipelines and the replay toolkit; generating the synthetic data sets that those simulations produce; and interpreting results. We have not used generative AI tools for translation, and the formulation and proof of mathematical claims and qualitative or thematic data analysis are not applicable to this work. Additionally, we used generative AI tools for the recommended-disclosure tasks of drafting parts of the paper, editing it for readability, suggesting its structure, creating and editing software code.

We have reviewed all AI-assisted work. The authors set the research questions, chose among the proposed designs, and take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work collects no new human data. The three human studies it draws on (Colley et al., 2025; Jansen et al., 2025; Susak et al., 2026) are published archival rating datasets, used here only to estimate within-rater noise and the parameters of the error processes (Section 4, Appendix B.4) and as fitted-oracle companions (Appendix B.5); no participant is identifiable from any quantity we report. The original publications state that two of the studies followed their university’s ethics-committee guidelines under local regulations that required no additional formal approval, and that the third received institutional ethics approval. In all three, participants gave informed consent, were compensated, and agreed to the anonymised publication of their data, which the original authors released openly. We use only those released ratings, within the terms of their publication. The analytic landscapes involve no human participants. We foresee no direct negative societal impact: the paper’s recommendations (confirm a claimed improvement before reporting it, and spend part of a human-feedback budget on identification) reduce the rate of false improvement claims rather than raise it.

REPRODUCIBILITY STATEMENT

Software and compute: BoTorch 0.18.2 on PyTorch 2.13 (a CUDA build, though the simulator uses the CPU only), Python 3.12 on Windows 11. A run costs about 21 worker-seconds and the main sweep about 470 worker-hours on a 24-worker machine. Code, run settings, and the analysis outputs that every table and figure is read from are in the anonymised repository at <https://anonymous.4open.science/r/hitl-mobo-robustness-tests-17E2/>.

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The appendix has six parts. Appendix A covers the controls, the one primary estimand and the intervals, the seeds of every arm, the landscapes, and the cells where noise appeared to help. Appendix B covers further checks on the headline, the currency of an error, the onset bound, the error processes in the archival data, the fitted-oracle design, the smooth subset, extra trials, three design choices, and the multi-objective arm. Appendix C covers the ship rules, the one-shot predictor and its pilot version. Appendix D covers the surrogate’s noise estimate, the incumbent, manipulated landscapes, and errors in the design rather than the rating. Appendix E covers the end-of-study procedures, the budget rule, their held-out validation, the budget-neutral and process arms, the remedies that cost no trials, comparisons, and change detection. Appendix F covers the void replication and its sensitivity analysis.

A DESIGN, CONTROLS AND ESTIMANDS

A.1 THE THREE CONTROLS

- **Model-free floor.** Random and Sobol never read an observation, so their excess search loss must be identically zero, and across all 12,800 such comparisons the largest absolute value is exactly zero. The analysis raises rather than warns if it is not, and the table reads the value from that check rather than printing a constant. This negative control shows that no observation noise reaches the candidate stream. The floors are then excluded from the robustness ranking, which a method that ignores data would otherwise win.
- **Headroom.** Where the clean run already reaches the optimum a null is guaranteed, not earned. The smallest clean-run advantage over the floor is 0.176 (Rosenbrock), so no landscape is ceiling-limited.
- **Uniform offset.** A constant added to every observation is invisible to a GP that standardises its targets and uses a posterior-mean incumbent. With a shared noise stream the two arms propose identical designs to below 10^{-9} for the initial design and three model-based candidates (asserted by test), after which floating-point differences are amplified; in the sweep, BIAS and GAUSSIAN from the first rating agree to within 0.012 at every magnitude.

A.2 ESTIMANDS, INTERVALS AND SEEDS

One estimand is primary: the deployed design’s excess regret at the final trial over its identically seeded clean twin, as a fraction of the landscape’s achievable improvement opt_z , averaged within a landscape over acquisitions and seeds and then over landscapes. Every other number is one of five derived quantities, named where it is used: (i) the final-trial search loss and (ii) the selection loss, the two terms of Equation 5, which sum to the primary estimand; (iii) the trajectory loss, the post-onset per-trial average of the search-loss excess, used for the acquisition ranking, the predictor and the currency test because it has lower variance, and bounded as Appendix B.3 shows; (iv) the share of the standard process’s cost of error that a process change recovers, $(R_{\text{noisy}}^{\text{std}} - R_{\text{noisy}}^{\text{adapt}})/(R_{\text{noisy}}^{\text{std}} - R_{\text{clean}}^{\text{std}})$, a ratio of landscape means (Appendix E.5); and (v) the gain over the standard process in units of the achievable improvement, the numerator of (iv), used where the cost in the denominator is near zero or pooled over magnitudes that differ by two decades, as for the k -curve.

Intervals. Intervals are percentile intervals from 2000 landscape resamples. With twenty clusters such an interval is known to run narrow, so we checked it by simulation, treating the empirical distribution of the twenty landscapes as the population and drawing 10,000 studies of twenty landscapes from it. For the selection share at 1σ , the shortlist recovery and the deployed loss at 1σ the percentile interval covers 92.2%, 92.9% and 92.7% of the time, BCa 92.1–94.0%, and a t interval with 19 degrees of freedom and a delta-method standard error 94.6–94.7%, at up to two points more width; for the selection share, with 20,000 resamples, it gives [32.7, 51.0] against the percentile [33.6, 50.2] (Section 5 prints the 2000-resample percentile interval, [34.1, 50.4]). The shortfall is width, not skew, which BCa cannot fix. The check cannot detect bias the empirical distribution shares, or dependence between landscapes, which share seeds by design. The currency test (Appendix B.2) is the only place the paper uses BCa intervals (Efron, 1987).

Seeds. Table 3 lists the seeds, landscapes and acquisitions of every arm. Procedures that change only the end of a run are replayed from the logs of the arm they modify and add no runs; wherever

a choice among such procedures is reported, it is made on seeds 7–11 and scored on seeds 12–16 (Appendix E.3).

Table 3: Seeds, landscapes and acquisitions of every simulated arm, with the number of runs. “Remedy arms” are the search-side and process arms of Tables 33 and 34.

arm	seeds	landscapes	acquisitions	runs
Main sweep	7–16	20	12	79,200
Preregistered replication	27–36	20	12	21,600
Budget $T = 25$ and $T = 100$	7–11	20	6	5,400 each
Instrument, known noise	7–11	20	6	5,400 each
Incumbent	7–11	20	7	6,300
Input error: slip, recorded slip, misclick	7–11	20	8	7,200 each
Manipulated landscapes (bump, Levy)	7–11	6	12	11,880
Multi-objective	7–11	7 problems	6	1,890
Halo arms (multi-objective)	7–16	4 problems	1	200 each
Manipulation ladder	7–16	3	12	3,240
Remedy arms	7–11	20	1–2	600–5,000 each
Fitted-oracle replication	7–16	3 datasets	2	540
Gross faults, capped scale, self-report	7–16	20	2	1,400 / 1,600 / 3,600
Self-report calibration, $\rho \in \{0, 0.3, 0.6\}$	7–11	20	2	600 each
Comparisons against ratings	7–9	20	1 each	960 fits
Fitted-oracle companion, two arms	7–16	3 datasets	12	3,240 each
Oracle isolation	7–11	20 datasets	6	2,400

A.3 THE LANDSCAPES AND THE DESCRIPTOR FIT

Table 4 lists the twenty landscapes with their measured geometry, and Table 5 places the three human studies of Section 4 on the same axis. Appendix C.2 reports the descriptor model only as a comparison for $\text{frag}(\sigma_e)$; Table 6 gives it in full, with the variance inflation factors that limit what can be read off it.

Table 4: The twenty landscapes and their measured geometry. opt_z is how many landscape standard deviations the optimum stands above a random design; sparsity is the fraction of the box within 10% of the optimum; ruggedness is 1 minus the lag-1 autocorrelation along random walks; tail is $\sigma/(1.4826 \text{ MAD})$; $\text{frag}(1)$ is the one-shot selection loss at an error of one landscape standard deviation. Names are the suite’s keys; `levy_10` has eleven inputs.

landscape	d	opt_z	sparsity	rugged.	skew	tail	$\text{frag}(1)$
powell	4	0.76	7.2e-01	0.13	−2.33	2.52	0.18
rosenbrock	4	1.03	4.9e-01	0.13	−1.06	1.13	0.18
branin	2	1.25	2.4e-01	0.13	−1.09	1.14	0.30
steering_law	4	1.29	7.9e-01	0.12	−3.33	1.44	0.43
power_law_practice	4	1.72	1.4e-01	0.13	−2.02	1.72	0.63
weber_fechner	4	2.15	5.5e-02	0.15	−0.71	1.01	0.64
levy_10	11	2.62	1.3e-02	0.20	−0.54	1.00	0.80
griewank	6	2.75	1.0e-02	0.12	−0.26	0.98	0.85
moving_peaks	4	2.96	1.8e-03	0.17	−0.53	1.04	0.64
hartmann_3	3	3.05	1.6e-02	0.14	+1.14	1.49	0.50
stevens	4	3.06	3.0e-03	0.12	−0.17	0.97	0.75
eggholder	2	3.21	6.6e-03	0.42	+0.04	1.00	0.68
yerkes_dodson	4	3.54	1.9e-03	0.12	+0.27	0.98	0.73
rastrigin	4	3.64	5.8e-04	0.58	−0.08	0.98	0.69
schwefel	4	4.33	1.2e-04	0.25	+0.00	0.99	0.69
hicks_law	4	5.40	<1.5e-05	0.12	+0.40	0.97	1.00
hartmann_6	6	7.97	9.2e-05	0.14	+2.67	2.74	0.53
michalewicz	5	8.05	<1.5e-05	0.36	+1.02	0.92	0.80
ackley	4	20.10	<1.5e-05	0.80	+3.26	1.67	0.18
shekel	4	56.77	<1.5e-05	0.13	+3.72	1.63	0.00

Table 5: Within-rater noise in three published human studies, in units of σ_f , the spread of the study’s fitted objective over its design box: a nearest-neighbour nugget with a participant-bootstrap interval, and the within-session residual SD, which bounds the noise from above. As logged, the `opticarvis` ratings mix two scales (Appendix B.4); as logged, `provoice` enters one lower-is-better construct with the wrong sign. The one-scale and sign-corrected rows are the ones the paper uses. The three σ_f are not one kind of surface: the `ehmi` oracle is fitted to per-design mean ratings and the other two to individual ratings; fitted to individual ratings, `ehmi`’s σ_f is 0.315 and its noise $0.86\sigma_f$ (`scripts/review_checks/ehmi_sigmaf.py`).

study	ratings	σ_f	noise / σ_f	upper bound
ehmi	as logged	0.365	0.74 [0.48, 0.97]	1.17
opticarvis	as logged, two scales mixed	0.374	3.35 [3.03, 3.58]	–
opticarvis	one scale	0.156	1.11 [0.81, 1.40]	1.34
provoice	as logged, one sign reversed	0.298	1.02 [0.20, 1.56] (weak)	2.58
provoice	sign corrected	0.449	0.60 [0.00, 0.90] (weak)	2.37

Table 6: Post-onset excess regret on landscape descriptors and the error magnitude, predictors z -scored, cluster-bootstrapped over landscapes with Benjamini–Hochberg correction (Benjamini & Hochberg, 1995). The variance inflation factor (VIF) is computed on the design matrix.

term	β	95% CI	p_{FDR}	VIF
$\log \sigma_e$	+0.37	[+0.15, +0.66]	0.001**	1.0
$\log \text{opt}_z$	+0.64	[+0.24, +0.74]	0.001**	9.2
$\log \text{tail}$	+0.20	[+0.03, +0.33]	0.008**	1.2
rugged.	−0.09	[−0.38, +0.12]	0.519	1.3
d	−0.03	[−0.26, +0.12]	0.768	1.0
$R^2 = 0.526$, 640 cells, 20 landscape clusters				

opt_z carries a VIF of 9.2 and regresses on the other descriptors with $R^2 = 0.89$, so its coefficient and those of sparsity and skew (excluded here, VIF 5.8 and 6.3) are not separately identified. The $\log \text{opt}_z$ point estimate also sits at the top of its own percentile interval, because one landscape carries it: leaving Shekel out ($\text{opt}_z = 56.8$) lowers the coefficient from 0.64 to 0.33, while leaving out any other landscape leaves it between 0.59 and 0.67. This is why Appendix D.3 builds landscapes that vary these properties independently rather than trying to disentangle them by regression. Dimension and ruggedness are null here. The fixed- opt_z ladder of Appendix D.3 shows dimension does raise the cost, so that null is a limit of the observational design.

A.4 DOES NOISE EVER HELP?

A noisy observation can in principle help, by pushing an over-exploiting optimizer out of a local basin. At one seed per cell, 25 of the 640 landscape-by-condition cells had negative mean excess regret with $d_z < -0.2$, noise apparently improving the outcome, which is the kind of finding that invites a mechanism.

At ten seeds it is one cell: `michalewicz` under AR(1) error at 0.05σ from trial 21, mean excess -0.136 , $d_z = -0.21$ over 100 paired runs. One cell in 640 is fewer than the fifteen or so that a null family of 640 tests would flag at this threshold. We report it as a null and note the sampling-variability lesson: the twenty-four cells that disappeared between one seed and ten had all shown noise helping.

B THE COST OF ERROR

B.1 THE CONCLUSIONS UNDER SEVEN FURTHER CHECKS

The surrogate’s noise estimate is not the problem. The GP’s fitted noise is an eighth of the injected value over the first fifteen observations at 1σ and above, yet giving it the true variance removes about 3% of the summed cost over the eight cells, -1% at 1σ from the first rating, and survives correction in no condition (Appendix D.1).

The incumbent is part of the method. The observed-max incumbent removes 20.0% of the excess-weighted cost, 24% at 1σ and 48–81% at 0.25σ and below, and worsens the deployed design: robustness rankings rank acquisition-and-incumbent pairs (Appendices D.2 and E.5).

opt_z is not causal on its own, and dimension is not inert. Purpose-built landscapes show that raising a narrow optimum never measurably raises the cost ($-0.001 [-0.035, +0.036]$ at 1σ), that a Levy ladder confounds dimension with signal strength, and that with opt_z fixed dimension still raises the cost at 1σ and 5σ , $1.7 \times [1.1, 5.3]$ and $1.7 \times [1.4, 2.3]$ from $d = 7$ to 11 (Appendix D.3).

Multi-objective BO is no more fragile. On six BoTorch problems the dose-response and onset structures replicate and the four hypervolume acquisitions are indistinguishable. Error from the first rating costs a seventh of the scalar suite’s at 0.05σ , a third at 0.25σ and a half at 1σ and 5σ , the intervals separating only at the two smallest (Appendix B.9).

The budget matters, through the onset bound. The rating scale cuts both ways. With the late onset held at 40% of the run, the onset ratio is $2.3 \times$, $5.1 \times$ and $11.7 \times$ at $T = 25, 50$ and 100 , and about $1 \times$ at each once divided by the clean run’s post-onset improvement; a discretised rating scale makes small errors costlier and large ones marginally cheaper (Appendices B.8 and B.3).

The replication is void; post hoc, the claims hold. Four preregistered claims ran on ten fresh seeds; one control failed, so the run is void by its own rule, and with that landscape excluded post hoc all four clear their thresholds (Appendix F.1).

The fitted-oracle design reports about a quarter of the deployed loss and a third of the trajectory loss on the same landscape. On three archival datasets it reports a third to a half of the cost the analytic arm measures from the first rating, but those are different landscapes. Holding the landscape fixed and fitting the oracle to synthetic archival data drawn from it, the fitted-oracle design reports 0.31 [0.20, 0.52] of the exact cost at 1σ on the trajectory (0.25 [0.14, 0.78] on the deployed design), less on 18 of 20 landscapes, and neither oracle fidelity nor the optimum estimate explains the gap (Appendix B.5).

B.2 WHAT MAKES AN ERROR BIG?

Standardising every objective assumes that a landscape’s own standard deviation is the right unit for its error. The rival is the improvement available, since a half-SD error is one thing when the optimum stands 0.76 SDs above a random design and another at 56.8. We fit the power law

$$\mathbb{E}[R] = A \sigma_e^{\beta_c} \text{opt}_z^{\beta_z} \quad (6)$$

to the landscape-by-magnitude cell means of the raw excess regret R , using quasi-Poisson estimating equations, the Poisson pseudo-maximum-likelihood estimator of a multiplicative mean (Gourieroux et al., 1984; Santos Silva & Tenreiro, 2006), which needs only the mean to be right and keeps the cells where noise helped (Appendix A.4). If the spread is the right unit (SPREAD), $\beta_z = 0$; if the error relative to the achievable gain is, $R = \text{opt}_z h(\sigma_e/\text{opt}_z)$, and for a power-law h that means $\beta_c + \beta_z = 1$ (GAIN). Intervals are BCa intervals over resampled landscapes, the unit of replication for a landscape-level covariate.

SPREAD fails at both onsets (Table 7): β_z excludes zero in all eight conditions, so standardising an objective does not make error magnitudes comparable, and an error magnitude reported without the landscape’s achievable improvement is under-specified. The cost also grows with opt_z faster

Table 7: Neither scale fits. Exponents of $\mathbb{E}[R] = A \sigma_e^{\beta_c} \text{opt}_z^{\beta_z}$ fitted to landscape $\times$ magnitude cell means, with 95% BCa intervals from 2000 landscape resamples and a jackknife over landscapes. β_z excludes zero in every condition; $\beta_c + \beta_z - 1$ excludes zero in six of eight.

error process	t_0	SPREAD: $\beta_z = 0$				GAIN: $\beta_c + \beta_z = 1$		verdict
		β_c	β_z	95% CI		$\beta_c + \beta_z - 1$	95% CI	
GAUSSIAN	0	0.51	0.86	[0.47, 1.13]		+0.37	[+0.15, +0.70]	neither
BIAS	0	0.51	0.85	[0.49, 1.08]		+0.36	[+0.17, +0.65]	neither
DRIFT	0	0.55	0.82	[0.55, 1.11]		+0.37	[−0.01, +0.63]	gain
AR(1)	0	0.66	0.85	[0.60, 1.15]		+0.51	[+0.14, +0.77]	neither
GAUSSIAN	20	0.62	1.15	[0.64, 1.48]		+0.76	[+0.03, +1.03]	neither
BIAS	20	0.70	1.14	[0.61, 1.58]		+0.85	[+0.04, +1.08]	neither
DRIFT	20	0.65	1.15	[0.61, 1.55]		+0.80	[−0.00, +1.14]	gain
AR(1)	20	0.71	1.14	[0.64, 1.49]		+0.85	[+0.13, +1.11]	neither

than GAIN allows: $\beta_c + \beta_z - 1$ is +0.36 to +0.51 from the first rating and +0.76 to +0.85 from trial 21, with intervals excluding zero in six of eight conditions, DRIFT touching it at both onsets; ordinary least squares on the logarithm of the positive cells rejects GAIN in all eight. The power law is itself an approximation: a centred $(\log \sigma_e)^2$ term is negative in all eight conditions and excludes zero in six, so β_c is an average slope over the four magnitudes. Neither verdict rests on it. With the four magnitudes given free levels, β_z is unchanged, since the panel is balanced. With h left free, a B-spline with four degrees of freedom in $\log(\sigma_e/\text{opt}_z)$, the coefficient that $\log \text{opt}_z$ adds to it is +0.31 to +0.44 from the first rating and +0.64 to +0.73 from trial 21 (BIAS, where four degrees of freedom do not converge, gives +0.57 with three), with percentile intervals excluding zero in all eight conditions, DRIFT included. The estimate leans on *shekel*, the one landscape with opt_z above 25; without it β_z is 1.04–1.10 and the gap 0.57–0.73 from the first rating. Appendix D.3 shows that opt_z is not what does the work: raising a narrow optimum does not raise the cost at any magnitude, and the dependence runs through how findable the optimum is.

B.3 THE ONSET BOUND

Let s be the last design a noisy run shares with its clean twin, $s = \max(t_0 + 1, 5)$: the design at trial $t_0 + 1$ is chosen from exact data, and the first five trials are a uniform initial design. Write $r_n(t)$ and $r_c(t)$ for the two runs’ search loss. A running maximum never gets worse, so for $t \geq s$

$$r_n(t) - r_c(t) \leq r_n(s) - r_c(t) = r_c(s) - r_c(t) = B(t),$$

the improvement the clean run makes after s . The bound is exact given the shared prefix, and the prefix is shared in all 73,600 model-based pairs of the main sweep and both budget arms: designs, true values and search loss are identical through s and observations through t_0 , and no pair exceeds its bound. It binds often after a late onset. At 1σ , 27% of late-onset runs make no progress after trial 21 while their clean twin does (7% after trial 5 at the early onset), and 43% of post-onset trials after a late onset with $B > 0$ sit exactly on it.

Divided by the bound, the late onset is not cheaper (Table 8): the normalised onset ratio is 0.95 [0.78, 1.20] at 1σ against the raw $5.0\times$, and the bound ratio, $5.3\times$, accounts for all of the raw one. The same holds for every process and at 0.25σ ($0.85\times$ [0.60, 1.25]), and at 5σ the late onset loses a larger share of what was left (75% [69, 80] against 63% [57, 69]); at 0.05σ the excesses are too close to zero for a ratio to mean anything. At 1σ , thirteen of the twenty landscapes lose a larger share after a late onset. The raw ratio invites the reading that an optimizer which has located a good region is hard to dislodge. The trajectory cannot support it: after a late onset the noisy run keeps what was banked by trial 21 by construction, and of what is still available it loses the same share an early-onset run loses of its own. The deployed design is not bounded this way, and in 61% of late-onset runs at 1σ its excess exceeds $B(T)$. There the late onset costs 13.6% [11.1, 16.0] against 29.7%, a ratio of $2.2\times$ [1.9, 2.6].

Table 8: The trajectory loss at 1σ and its bound, in percent of the achievable improvement, with landscape-bootstrap intervals. The last column is the excess as a share of the bound (a ratio of landscape means). B depends only on the clean run, so it is the same for every error process.

error	onset	excess	bound B	excess / B
GAUSSIAN	trial 1	14.8 [11.5, 18.3]	34.6	43 [38, 48]
	trial 21	2.7 [1.7, 3.8]	6.6	41 [31, 51]
BIAS	trial 1	14.7 [11.5, 18.2]	34.6	43 [37, 48]
	trial 21	3.2 [2.0, 4.5]	6.6	49 [41, 58]
DRIFT	trial 1	15.2 [11.7, 18.9]	34.6	44 [39, 49]
	trial 21	2.9 [1.8, 4.1]	6.6	44 [37, 52]
AR(1)	trial 1	11.5 [9.1, 14.1]	34.6	33 [30, 37]
	trial 21	2.4 [1.5, 3.4]	6.6	37 [28, 46]
pooled	trial 1	14.1 [10.9, 17.4]	34.6 [28.8, 40.9]	41 [36, 45]
	trial 21	2.8 [1.8, 3.9]	6.6 [4.0, 9.4]	43 [34, 51]

B.4 THE ERROR PROCESSES IN THE ARCHIVAL DATA

The error processes of Section 3.2 are parameterisations chosen before looking at human data. The three archival studies contain enough repeated structure to check them: several ratings per participant in one session, designs that recur, and in `provoice` a counterbalanced crossover in which the same design is rated in both conditions of one sitting. A rating is modelled as the design’s value plus the rater’s error. The design’s value is removed with a leave-one-participant-out model of rating on design, of the family the fitted-oracle pipeline selects, so that a participant’s own ratings cannot fit their own residuals away, and every interval resamples participants (2000 draws). Table 9 sets the estimates beside what the simulation assumes.

Table 9: Error-process parameters estimated from three archival studies, with participant-bootstrap intervals, beside what the simulation assumes. Noise, drift and offsets are in units of the study’s σ_f or of its noise SD as stated. The simulation’s lag-one entry is what the same estimator reads under the paper’s AR(1) at these session lengths. Drift and serial dependence are weakly identified (see text). `opticarvis` is on its one-scale ratings, in units of the σ_f of the oracle family selected on the as-logged data (0.258; the family selected on the corrected data gives 0.156, Table 5); the `provoice` column is its sign-corrected composite, and its drift the crossover estimate.

quantity	ehmi	opticarvis	provoice	simulation
noise / σ_f (nugget)	0.74 [0.48, 0.97]	0.67 [0.49, 0.85]	0.60 [0.00, 0.90]	swept
lag-one autocorrelation	0.44 [0.27, 0.64]	0.27 [0.11, 0.43]	0.61 [0.38, 0.79]	0.81–0.82
session trend removed	0.10 [−0.04, 0.28]	−0.22 [−0.31, −0.12]	0.05 [−0.10, 0.16]	0.25–0.51
drift over a session / σ_f	1.21 [0.40, 1.61]	0.40 [0.18, 0.89]	0.74 [−0.36, 1.63]	σ_e
drift, between-rater SD / σ_f	1.59 [0.89, 2.15]	1.06 [0.61, 1.36]	3.75 [2.59, 4.73]	0
rater offset SD / noise SD	0.97 [0.73, 1.13]	1.42 [1.03, 1.82]	0.71 [0.51, 0.82]	1
residual SD, best / worst third	0.45 [0.30, 0.62]	0.61 [0.47, 0.85]	0.74 [0.61, 0.90]	1
residual excess kurtosis	5.2 [2.8, 7.8]	9.1 [3.3, 15.1]	2.7 [0.9, 5.9]	0
ratings at the top of a scale, %	83 [74, 91]	55 [43, 66]	63 [48, 77]	0

Two defects in the fitted-oracle pipeline’s composites. In `opticarvis`, 40 of the 586 ratings were logged on the raw instrument scales (1–5, 1–7 and −3 to 3) and the rest on the $[-1, 1]$ scale the optimizer saw. Averaged together, they make one design a spike about four times the height of the rest of the surface, which sets that oracle’s optimum and its held-out R^2 and puts its noise at $3.35\sigma_f$ (the as-logged row of Table 5); on the scale-consistent ratings the noise is 0.67 [0.49, 0.85] with the oracle family selected on the as-logged data and 1.11 with the one selected on the corrected data, and no model family reaches a positive held-out R^2 . In `provoice` one lower-is-better construct entered the composite with the wrong sign. The corrected data drop the off-scale rows and enter the construct with a minus sign, and the fitted-oracle companion arm runs on the corrected data. Both defects touch only the fitted-oracle companion arm of Appendix B.5 and the noise anchor; the analytic arm and the oracle-isolation experiment never read these data.

What the data show. The design model predicts held-out participants poorly (leave-one-participant-out R^2 0.23, -0.21 and 0.08 ; the pipeline’s 0.55 for `ehmi` comes from folds that let a participant’s own ratings into training), and search progress rises with trial order, so drift and serial dependence cannot be fully separated from progress the model misses; the one estimate free of that confound is `provoice`’s crossover, which gives the drift entry of its column. With that caveat, the four main processes fare differently. GAUSSIAN noise of 0.6 to $1.1\sigma_f$ is what the nugget measures in the pipeline’s σ_f (Table 5), though the nugget can understate the noise, and the within-session residual SD bounds it from above at 1.17 , 1.34 and $2.37\sigma_f$ (0.81 for `opticarvis` in the units of Table 9). It can also overstate it, since close designs are not replicates: halving the closeness radius lowers the `ehmi` nugget by 11% and doubling it raises it by 39% . BIAS matches in size: raters differ by offsets of 0.7 to 1.4 noise SDs, though whether an offset shared by every rater exists cannot be identified from ratings alone. DRIFT matches in size but not in form: sessions drift by 0.4 to $1.2\sigma_f$, but in both directions and by amounts that differ between raters, rather than on one common ramp. AR(1) with $\rho = 0.8$ does not match: the raw lag-one autocorrelation lies below what the same estimator reads under it in every study, and once each session’s own trend is removed it is near zero in two studies and negative in `opticarvis`, so most of the apparent serial dependence is rater-specific drift. An $\rho = 0.8$ component carrying a minority of the residual variance is not excluded (a third of it in `ehmi`, with an interval reaching 0.62), and the AR(1) arm is best read as a stress test. The data also show two faults the main arms omit: tails heavier than Gaussian (1.7 to 2.4% of residuals beyond three SDs, against 0.3%), which the gross-fault arms model, and saturation, with at least one construct at its best end in 55 to 83% of ratings and noise that shrinks for the best designs as a bounded scale compresses them. The main arms have no bound; the capped-scale arm of Appendix E.4, among the most expensive faults measured, is the realistic one.

B.5 WHAT THE FITTED-ORACLE DESIGN WOULD HAVE CONCLUDED

The introduction argued that fitting a regression oracle to archival ratings entangles the cost of error with the cost of the oracle being wrong. To measure this rather than argue it, we run the data-driven simulator on the three archival datasets at the same error magnitudes, expressed in each dataset’s own σ_f so that “ 1σ ” means the same thing in both arms.

Table 10: The same experiment, run against exact objectives and against oracles fitted to ratings, under GAUSSIAN error. Percentages are post-onset excess as a fraction of the gap between the optimum and a same-budget model-free floor (the floor gap), the one normaliser computable in both arms. From the first rating the fitted-oracle design reports a third to a half of the cost.

arm	error from trial 1				error from trial 21			
	0.05σ	0.25σ	1σ	5σ	0.05σ	0.25σ	1σ	5σ
known functions (20)	10	44	99	148	3	7	10	16
fitted oracle (3)	4	15	46	68	0	4	10	10
fitted, no augmentation (3)	3	17	46	64	2	7	9	10

The fitted-oracle arm has no verified optimum, so Table 10’s denominator uses the best value any arm reached, which is attainable by construction and so inflates the fitted fraction: the bias works against the result below. Three design spaces against twenty also widen the intervals. Two of the three datasets carried the logging defects of Appendix B.4; the arm is run here on the fixed data, on which neither of those two oracles predicts held-out ratings (cross-validated R^2 -0.09 and -0.07).

Across different landscapes, the fitted-oracle design reports a third to a half of the cost, and only from the first rating. At 1σ from the first rating the analytic arm puts the loss at 99% of the floor gap (cluster-bootstrap interval $[54, 154]$) and the fitted-oracle arm at 46% ($[31, 75]$), a ratio of 2.2 , interval $[1.0, 4.4]$ from resampling the twenty landscapes and the three datasets independently; three datasets allow only ten distinct resamples. At 0.05σ the fitted-oracle arm reports 4% ($[1, 7]$) against 10% ($[4, 19]$). From trial 21 the gap closes: the intervals overlap at every magnitude, and at 1σ the two arms agree (10% against 10%). On the as-logged data the arm reports a third of the cost, the two affected datasets at 5% and 19% of the floor gap against 31% each on the corrected data.

On the corrected data the augmentation contrast does not test fidelity. There, turning off the jitter augmentation changes the oracles' cross-validated R^2 by at most 0.03 (on the as-logged data it raised one by 0.29), and rerunning the arm without it moves the fraction destroyed by $+0.017$ $[-0.003, +0.037]$ on `ehmi`, $+0.006$ $[-0.006, +0.018]$ on `opticarvis` and -0.015 $[-0.056, +0.020]$ on `provoice` (800 paired cells each, seed-bootstrap intervals). The isolation experiment below is the test of fidelity. The optimum estimate is not the cause of the gap either: it cancels in the excess, a difference of two regrets, and enters only the denominator, which biases the other way. The three fitted design spaces sit in the lower two thirds of the twenty-landscape spread (per-landscape fractions at 1σ run from 0.11 to 4.8; the fitted three are 0.31, 0.31 and 0.75, above six, six and thirteen of the twenty), and with three design spaces the arm cannot say whether that is a property of human objectives or of these studies.

Holding the landscape fixed. Set against twenty analytic landscapes, three human objectives cannot say whether the gap above belongs to the oracle or to the landscapes. The isolation experiment (`scripts/oracle_isolation.py`) removes that confound. For each analytic landscape it builds a synthetic archival dataset the way one arises: 37 participants, the size of `ehmi`, each contributing the first 20 designs of a clean BO run on that landscape, so that the designs cluster where BO goes, rated once with GAUSSIAN noise at $1\sigma_f$. It then runs the companion arm's pipeline on it unchanged: the oracle family chosen by the same grouped cross-validation over seven model families (gradient boosting for fifteen landscapes, extra trees for three, CatBoost for two), fitted with the same augmentation, the box taken from the data, the optimum estimated by the best value any clean run reached, and the error grid scaled to the fitted oracle's own σ_f (0.78 to 1.21). The exact arm is the main sweep on the same landscape, acquisitions (LogEI, qNEI, UCB, EI), seeds 7–11 and onset, and both use the floor-gap fraction of Table 10.

Table 11: Holding the landscape fixed: the fitted-oracle design against the exact objective on the same twenty landscapes, under GAUSSIAN error from the first rating. Percentages are excess regret as a fraction of the floor gap, as in Table 10, which is small on some landscapes, so deployed values can exceed 100%; the ratio is of landscape means, with a landscape-bootstrap interval; p is a Wilcoxon signed-rank test over the twenty landscapes.

error	fitted-oracle design	exact objective	ratio	fitted lower	p
<i>search loss, post-onset trajectory</i>					
0.25 σ	1.8%	14.4%	0.13 $[-0.15, 0.36]$	16/20	0.0006
1 σ	17.7%	57.9%	0.31 $[0.20, 0.52]$	18/20	0.0003
5 σ	39.4%	112.7%	0.35 $[0.22, 0.59]$	16/20	0.0006
<i>deployed design, final trial</i>					
0.25 σ	11.2%	76.0%	0.15 $[0.08, 0.53]$	14/20	0.006
1 σ	65.1%	255.6%	0.25 $[0.14, 0.78]$	15/20	0.014
5 σ	164.8%	872.6%	0.19 $[0.09, 0.56]$	14/20	0.019

On the same landscape the fitted-oracle design reports a third of the trajectory loss at 1σ and 5σ and an eighth at 0.25σ , and a seventh to a quarter of the deployed loss, less on 14 to 18 of the twenty landscapes (Table 11); on the six laws alone it reports 19.8% against 40.3% at 1σ . The oracles are not poor: they correlate 0.45 to 0.94 with the true landscape at fresh points (median 0.84). Their fidelity does not predict the trajectory ratio at 1σ (Spearman -0.13), and at 5σ and on the deployed design the better oracles understate more (-0.55 at 5σ on the trajectory, -0.50 at 1σ on the deployed design). The fitted-oracle design does not preserve the ranking of landscapes either: its per-landscape cost correlates 0.11 with the exact one at 1σ . The estimated optimum again biases the fitted fraction upward, against the finding. A likely mechanism, which we have not isolated, is that a tree-ensemble oracle is flat where the archival designs are dense, so that near its optimum many designs tie and a wrong pick costs little.

The fitted-oracle design does not preserve the acquisition ranking either. Within every landscape and magnitude the four acquisitions run in both arms (EI, LogEI, UCB, qNEI) were ranked by trajectory loss, in the fitted-oracle arm on seeds 7–11 and in the exact arm on seeds 7–16, both under GAUSSIAN error from the first rating (`scripts/oracle_isolation_acq_ranking.py`); ranks are taken within a cell, so neither arm's normaliser matters. Pooled over the three magnitudes the exact arm's mean ranks are 2.2 (qNEI), 2.2 (UCB), 2.8 (EI) and 2.9 (LogEI); the fitted-oracle

arm’s are 2.7, 2.6, 2.5 and 2.3, a smaller spread in the other direction. The most robust acquisition of a landscape agrees between the arms in 3, 8 and 2 of 20 cells at 0.25σ , 1σ and 5σ (chance is 5). Four acquisitions and twenty landscapes make this descriptive, and the two arms differ in their seed sets, but a study that ranks acquisitions on such an oracle would have ranked these four the wrong way round.

B.6 THE SMOOTH SUBSET, LANDSCAPE BY LANDSCAPE, AND THE SURROGATE’S FIT

The subset was written down before any result was computed, from the landscape definitions alone: the six human-performance laws, of which Yerkes–Dodson and the steering law have one interior peak and Stevens, Hick, Weber–Fechner and the power law of practice are monotone with a corner optimum. Four of them (Yerkes–Dodson, Stevens, Hick, Weber–Fechner) apply the one-dimensional law to each of four inputs and average, so they are additive across inputs; the power law of practice and the steering law map three inputs to the law’s parameters and the fourth to its argument. They differ from the classical functions in more than smoothness. All six are four-dimensional, and four of them have their optimum at a corner, which among the classical functions only Eggholder shares, on one face. A secondary split, also fixed in advance, separates the three classical functions that are convex or effectively unimodal on their box (Powell, Rosenbrock, Griewank) from the other ten.

Table 12: The deployed design’s excess by group, percent of the achievable improvement, pooled over the four error processes, with landscape-bootstrap intervals, and the selection share of the first column.

group	n	deployed excess			selection
		1σ , trial 1	1σ , trial 21	5σ , trial 1	share
all landscapes	20	29.7 [27, 33]	13.6 [11, 16]	71.1 [64, 78]	42%
human-performance laws	6	32.7 [29, 37]	9.5 [6, 13]	71.5 [66, 79]	40%
classical functions	13	27.8 [24, 32]	14.8 [12, 17]	70.4 [61, 80]	42%
convex or unimodal	3	27.2 [18, 39]	14.6 [10, 19]	79.2 [67, 101]	62%
rugged	10	28.0 [24, 33]	14.8 [12, 18]	67.7 [57, 79]	36%

The laws lose as much as the classical functions from the first rating (Table 12; a difference of +4.8 [−1, +11] points at 1σ and +1.1 [−11, +13] at 5σ) and less when the error arrives late (−5.2 [−10, −1] at 1σ), where more of their cost is selection (87% against 65%). Nothing in the headline is produced by the multimodal functions. Figure 2 gives the headline cell per landscape: the deployed loss has a median of 29.0% and runs from 18.3% (Rosenbrock) to 46.6% (Hartmann-6), and the selection share from 4% (Shekel, where the optimum is so narrow that the search rarely reaches it) to 85% (Powell); it exceeds one half on seven landscapes.

Ship rules on the subset. Under GAUSSIAN error at 0.25σ and above, the posterior-mean maximiser recovers +11.5% [6, 16] of the deployed loss on the laws, beating the best observed rating on all six, and −3.3% [−10, +2] on the classical functions; the three convex or unimodal classical functions behave like the laws (+9.2% [2, 17]) and the rugged ten do not (−7.4% [−14, −1]). The cautious rules hold up on both: the posterior mean less two latent SDs recovers +13.9% [11, 17] on the laws and +8.8% [1, 16] on the classical functions.

The surrogate at $n = 50$. We refitted the loop’s own GP on 800 runs, forty per landscape, half clean and half under 1σ GAUSSIAN error from the first rating. The noise estimate has recovered in both groups (0.84 and 0.85 of the injected SD), and the posterior mean does shrink the ratings everywhere: its error at the visited designs is 0.52 of the raw ratings’ on the laws and 0.67 on the classical functions. What separates the groups is the fit itself, and it shows with no error at all. On clean runs the fitted noise SD is 0.03 on the laws and 0.30 on the rugged ten, where the true noise is zero, so the GP reads their roughness as noise; the geometric-mean lengthscale is 0.85 against 0.34 box widths; and R^2 at 256 fresh Sobol points is 0.67 against −0.24. Across landscapes the GP’s leave-one-out predictive fit correlates with the posterior-mean maximiser’s gain at Spearman 0.62, and 0.85 within the classical functions. Within a landscape the run-to-run fit does not predict it (0.07), so a per-run fit check would not tell a practitioner when to trust the posterior. Structured

error misleads the posterior even where it fits: at 5σ from trial 21 the rule recovers -15.9% of the cost under DRIFT on the laws, and -9.6% $[-22, +2]$ under BIAS.

B.7 EXTRA TRIALS, BY ARM

For each noisy run we count the trials it needs to come within 1% of the achievable improvement of its clean twin after k trials, on the budget-100 arm of Appendix B.8 (GAUSSIAN error, six acquisitions; Table 13; every arm in Table 14). Against a clean 25-trial study, error of 0.05σ from the first rating costs a median of one extra trial and 0.25σ seven; 1σ costs a median of 41, the noisy run reaching that regret at a median trial of 56, 2.2 times the clean study’s, and 39% of runs never get there within 100 trials; at 5σ , 70% never do. Error from trial 41 cannot delay a run whose clean twin was already within 1% of its trial-50 regret by trial 41, and 84% of runs (505 of 600) were, so the median of zero in the right column is structural. In the other 16%, error from trial 41 costs a median of 8 extra trials at 0.25σ and 50 at 1σ , and 49% of them never recover within 100 trials at 1σ , 78% at 5σ ; over all runs that is the 8–12% in the table.

Table 13: Extra trials a noisy run needs to reach the regret its clean twin had after k trials, within 1% of the achievable improvement (budget-100 arm, GAUSSIAN error, six acquisitions): median over runs, and the share that never got there within 100 trials.

error	error from trial 1, match a clean 25-trial study	error from trial 41, match a clean 50-trial study
0.05σ	1 (4%)	0 (1%)
0.25σ	7 (14%)	0 (3%)
1σ	41 (39%)	0 (8%)
5σ	>78 (70%)	0 (12%)

Table 14 repeats the count of Table 13 for every arm at the early onset, against a clean ten-trial study within the fifty-trial budget, with the same one-percent tolerance. Fewer than half the runs are censored in every cell, so every median is exact. The strict count with no tolerance is in `extra_runs.csv` beside each arm. The recorded slip costs fewer trials than the unnoticed one at every magnitude above 1%, and a misclick almost none below 40% of the trials.

Table 14: Median extra trials to match a clean ten-trial study within 1% of the achievable improvement, error from the first trial, with the share of runs that never do within 50 trials. Response-error rows are the main sweep over all ten model-based acquisitions and ten seeds; input-error rows are six acquisitions and five seeds. In 18% of runs (369 of 2,000 clean twins) the clean run reached its trial-10 regret within the five shared initial designs, so the error could not delay them; without them the GAUSSIAN medians at 0.25σ and 1σ are 3 and 17, and at 5σ 54% never match.

error	0.05σ	0.25σ	1σ	5σ
GAUSSIAN	0 (1%)	1 (8%)	9 (26%)	26 (44%)
BIAS	0 (1%)	2 (9%)	8 (27%)	28 (45%)
DRIFT	0 (1%)	2 (9%)	8 (28%)	36 (49%)
AR(1)	0 (1%)	1 (4%)	3 (22%)	23 (42%)
error	1%	5%	15%	40%
SLIP	0 (3%)	3 (7%)	8 (15%)	24 (40%)
SLIP, logged	0 (2%)	2 (5%)	4 (10%)	14 (25%)
MISCLICK	0 (1%)	0 (3%)	0 (9%)	6 (23%)

B.8 THREE DESIGN CHOICES THAT COULD HAVE PRODUCED THE RESULT

Every number above was measured at a fixed budget, against an unbounded continuous response, using standard acquisition functions. Each of those is a choice, and each could be doing the work. The three arms here ran GAUSSIAN error on seeds 7–11 with six acquisitions (LogEI, EI, PI, UCB,

qUCB, qNEI) or, for the robust baselines, the two named; every reference row is the main sweep restricted to the same error process, acquisitions and seeds.

Table 15: Budget, with the late onset held at 40% of the run in each case and the reference row restricted to the arm’s own error process, acquisitions and seeds, so that budget is the only thing that changes. Percentages are the fraction of the achievable improvement destroyed.

	budget	error present	0.05σ	0.25σ	1σ	5σ
$T = 25$	from trial 1		0.6	3.2	9.4	15.4
	from trial 11		0.3	1.3	4.1	8.0
	ratio		$2.2\times$	$2.4\times$	$2.3\times$	$1.9\times$
$T = 50$	from trial 1		0.6	3.8	11.7	21.3
	from trial 21		0.1	1.0	2.3	4.7
	ratio		$4.4\times$	$3.9\times$	$5.1\times$	$4.6\times$
$T = 100$	from trial 1		0.2	3.4	11.1	23.9
	from trial 41		0.1	0.4	0.9	2.2
	ratio		$1.4\times$	$9.2\times$	$11.7\times$	$10.8\times$

Budget, and why the onset effect grows with it. Re-running at $T = 25$ and $T = 100$ with the onset held at the same fraction of the run gives Table 15. Cost from the first rating is roughly flat in budget (9.4%, 11.7%, 11.1% at 1σ), but cost from the late onset on the trajectory falls steeply, from 4.1% at $T = 25$ to 2.3% at $T = 50$ to 0.9% at $T = 100$, so the onset ratio grows from $2.3\times$ to $5.1\times$ to $11.7\times$. The rescaling that lets one budget speak for another is the bound of Appendix B.3: the improvement the clean run still makes after the onset shrinks from 10.2% to 6.3% to 3.0% of the achievable improvement as the budget grows, and divided by it the onset ratio is $1.0\times [0.9, 1.2]$, $1.0\times [0.7, 1.3]$ and $0.9\times [0.5, 3.4]$. The budget changes how much improvement is left when the error arrives, not what share of it the error destroys.

Table 16: A bounded, discrete response. The instrument arm clips to $\pm 8\sigma$ and rounds to 0.55σ , the range and step implied by the three archival studies. Both rows: GAUSSIAN error, six acquisitions, five seeds. Only the noisy run’s post-onset ratings are clipped and rounded; its clean twin and every earlier rating are exact and continuous, so each bounded row measures noise and quantisation together, and the from-trial-21 row measures an instrument that switches on at trial 21.

response scale	error present	0.05σ	0.25σ	1σ	5σ
unbounded, continuous	from trial 1	0.6	3.8	11.7	21.3
	from trial 21	0.1	1.0	2.3	4.7
bounded, discrete	from trial 1	3.8	5.0	11.7	20.8
	from trial 21	1.3	1.4	2.5	4.6

A real rating scale, which cuts both ways. The synthetic objective is unbounded and continuous; a person returns a bounded, discrete rating. The three archival instruments have steps of 0.55 , 0.54 and 1.12σ and span sixteen to twenty σ , so the arm clips to $\pm 8\sigma$ and rounds to 0.55σ , the two five-item instruments, in the study’s own currency. Table 16 shows the two effects separately, against a clean twin that is not discretised; we read only the rows for error from the first rating, since from trial 21 the scale itself arrives mid-run. At 0.05σ the discretised arm is six times worse (3.8% against 0.6%), because rounding to a 0.55σ grid is itself an error of about 0.16σ and swamps an injected error far smaller than the step; at 0.25σ it is still worse (5.0% against 3.8%); at 1σ the two are indistinguishable (11.7% against 11.7%); and at 5σ it is slightly better (20.8% against 21.3%), because clipping bounds how far one bad rating can move the incumbent. A rating scale is a noise floor at the bottom of the range and, at the top, a marginal safety rail.

Methods built for noise do not recover the cost. All ten model-based acquisitions above are ordinary ones run under noise. Two baselines built for it: the knowledge gradient, which values the improvement in the posterior *maximum* rather than in any observed value, and spending every

Table 17: Robust baselines. qKG is the batch knowledge gradient (Wu & Frazier, 2016) in BoTorch’s one-shot form (Balandat et al., 2020); replication spends every second evaluation re-asking about the incumbent. Compared at a fixed number of evaluations, so replication buys averaging with half the designs. The standard row is the ten model-based acquisitions of the main sweep on the same error process and seeds.

acquisitions	error present	0.05σ	0.25σ	1σ	5σ
ten standard	from trial 1	1.1	5.6	13.9	21.5
	from trial 21	0.3	1.3	2.7	4.5
qKG and replication	from trial 1	0.3	2.2	9.2	16.2
	from trial 21	0.2	0.1	2.2	4.8

second trial re-asking about the incumbent. Scored the way the rest of this appendix is, against each method’s own clean run, they lose less: 9.2% of the achievable improvement at 1σ from the first rating against the standard acquisitions’ 13.9%, and 16.2% against 21.5% at 5σ (Table 17). That comparison flatters them. Replication re-rates the incumbent in its clean run too, which carries no information, and qKG’s clean run is itself worse than the standard acquisitions’, so part of the smaller excess is a worse reference. Scored against the standard process’s clean run (Appendix E.5), replication loses 114% of the cost on the trajectory and recovers nothing on the deployed design (-12% [$-31, 5$]), and qKG loses 50% and 39%, recovering some of it only at 5σ from the first rating ($+21\%$ on the trajectory; pooled figures in Table 34). What remains is the information the error destroys, which is what Equation 7 measures.

B.9 DOES ANY OF THIS HOLD FOR MULTIPLE OBJECTIVES?

The applied setting is frequently multi-objective (Daulton et al., 2021) and everything above is scalar, so the generalisation cannot be assumed. We repeat the design on seven BoTorch problems with a published maximum hypervolume (BraninCurrin, ZDT1, DTLZ2, DH2, VehicleSafety, Penicillin and CarSideImpact), spanning two to four objectives and two to seven dimensions, with the four hypervolume acquisitions (qEHVI, qNEHVI and their log variants) and the same two model-free floors, under GAUSSIAN error, at five seeds: 1,680 paired runs, of which 960 are model-based noisy-versus-clean comparisons on the problems that survive the headroom screen below.

Standardisation transfers exactly rather than approximately. Each objective is mapped to zero mean and unit variance over a fixed scrambled-Sobol sample of 2^{15} points, and under a per-objective affine map the hypervolume scales by $\prod_m \sigma_m$, so the standardised maximum follows in closed form from the published one. We verify this rather than assert it: across the seven problems the largest relative error between the closed-form standardised maximum and a direct recomputation is 4.7×10^{-16} .

Controls. The model-free floor is again identically zero across all 560 such comparisons, so no observation noise leaks into the candidate stream in the hypervolume code path either. No run beats its published maximum hypervolume. Acquisition-optimization fallbacks total zero, which we report rather than assume: a hypervolume acquisition that fails to construct falls back to random sampling, and because that produces a plausible-looking curve rather than an error, the arm has to be read off the fallback counter and not off the plots.

One problem is dropped, by the same screen as before. DH2’s reference point leaves a 32,768-point Sobol sample with a hypervolume of exactly zero, and its best learner does not improve on the model-free floor at all. There is nothing for error to destroy, so its null is guaranteed by construction. It is excluded. The remaining six pass, with Penicillin weakest at 0.197, comparable to Rosenbrock at 0.176, the weakest admissible scalar landscape.

The comparison needs one piece of care. Elsewhere the paper divides excess regret by opt_z , a z-score of the optimum against a random design’s mean. A Pareto problem has no scalar optimum and therefore no opt_z , and the natural hypervolume analogue, the gap between the published maximum and what a model-free design of the same budget reaches, is a numerically much smaller denominator. Quoting one arm’s published fraction against the other’s would manufacture a threefold

Table 18: The two arms on a common footing. Both columns are post-onset excess divided by the gap between the optimum and a same-budget model-free floor, as a percentage; brackets are 95% cluster-bootstrap intervals over landscapes/problems. This is *not* the normalisation used in Table 1, and the two must not be read against each other. The scalar column is recomputed from the same runs as the known-function row of Table 10, under GAUSSIAN error.

error	scalar (20 landscapes)		multi-objective (6 problems)	
	from trial 1	from trial 21	from trial 1	from trial 21
0.05σ	10.2 [4, 19]	2.8 [1, 5]	1.4 [0, 3]	0.5 [-1, 2]
0.25σ	44.3 [22, 71]	6.5 [4, 10]	13.6 [8, 19]	3.7 [0, 10]
1σ	99.0 [54, 154]	10.3 [7, 13]	49.1 [42, 56]	10.7 [2, 24]
5σ	148.4 [76, 239]	16.5 [11, 22]	72.3 [55, 91]	13.0 [3, 27]

difference out of the choice of divisor alone. Both columns of Table 18 are therefore recomputed on the floor denominator, identically; and we also report the onset ratio, which is a ratio of two costs on the same problem and so does not depend on the denominator at all.

The two headline structures replicate. Cost rises monotonically with error magnitude and falls sharply when the error arrives late, on both arms. At 1σ from the first rating the multi-objective arm loses 49.1% of the hypervolume floor gap [42, 56] against the scalar arm’s 99.0% [54, 154]; arriving at trial 21 the two are indistinguishable, 10.7% [2, 24] against 10.3% [7, 13].

There is no evidence multi-objective BO is the more fragile of the two, and some that it is less. The point estimates for error from the first rating are half the scalar arm’s or less at every magnitude, a seventh at 0.05σ , a third at 0.25σ and half at 1σ and 5σ , and the intervals separate at 0.05σ and 0.25σ ; at 1σ they overlap slightly and at 5σ substantially, so this is a consistent direction rather than a clean separation at the top of the range. The onset ratio is correspondingly flatter: 2.8–5.6 $\times$ across magnitudes against the scalar arm’s 3.6–9.6 $\times$. A mechanism consistent with both observations, and with Appendix D.2, is that a hypervolume objective has no single incumbent to corrupt, so the *search* keeps finding good designs. The front the study would *report* is another matter: at 1σ from the first rating about half of it is truly dominated (5.1 of 10.9 designs), it holds 29% of the truly non-dominated designs evaluated, and its hypervolume shortfall exceeds that of the evaluated set in all eight cells.

Table 19: Hypervolume acquisitions, ranked over problems within each condition. Absolute loss is the fraction of the hypervolume floor gap lost under noise. The four are not distinguishable.

acquisition	mean rank	absolute loss
qNEHVI	2.40	20.2%
qLogEHVI	2.44	20.4%
qEHVI	2.56	20.7%
qLogNEHVI	2.60	20.8%

The choice among hypervolume acquisitions does not appear to matter. Mean ranks span 2.40 to 2.60 on a scale where 2.5 is a tie, absolute losses span 20.2% to 20.8%, and the Friedman test reaches $p < 0.05$ in 0 of 8 conditions (Table 19). Kendall’s W over condition-averaged losses is 0.056, below the $1/6$ that six problems give with no agreement at all, and each of the four acquisitions is the most robust on at least one of the six problems. Computed the same way, the scalar suite gives 0.52 over its ten acquisitions but 0.105 over four of the expected-improvement family under GAUSSIAN error, so most of that contrast is the number and spread of the acquisitions compared. Six problems at five seeds is far less power than twenty at ten, so this is “no detectable difference” rather than “no difference”. It does, though, point the same way as Section 6: the design spaces do not agree on which acquisition is most robust, though four closely related acquisitions on six problems cannot say whether they disagree more than the scalar suite does.

What does not transfer. Equation 7 is defined for a scalar objective: it is the loss from one greedy pick among m candidates, and on a Pareto problem there is no single greedy pick whose

loss it measures. Appendix C.2’s result is therefore untested in this arm rather than confirmed or refuted by it. A hypervolume analogue would have to define the one-shot loss over hypervolume contributions and take a stand on how per-objective error propagates into them. Table 20 is not claimed to extend to it.

C SEARCH, SELECTION AND PREDICTION

C.1 RE-SCORING THE DESIGN A STUDY SHIPS

The deployed regret everywhere else in this paper uses one ship rule: the design with the single best observed rating. To separate search from selection (Section 5) we re-score what the same trials would have shipped under other rules, adding no trial. The loop’s own surrogate, BoTorch’s default single-task GP, normalised over the landscape’s box and standardised, is refitted on every logged observation, and each rule picks among the *visited* designs: the best observed rating (the standard rule), the best mean rating, the posterior-mean maximiser, the maximiser of the posterior mean less one or two latent SDs, and, as the ceiling, the visited design with the best true value. The arm’s own logged rule must reproduce the logged final regret to 10^{-9} in every run or the analysis stops; over the 66,000 runs it did, with no failed fits.

Scored as a share of the deployed loss of GAUSSIAN error, the cautious rule (posterior mean less two latent SDs) recovers +5.8% [2, 10] at 1σ from the first rating, +12.6% [4, 23] at 1σ from trial 21, +9.7% [5, 15] and +18.5% [11, 26] at 5σ . It pays a fixed price of 0.015 of the achievable improvement whether or not error is present, so below 0.25σ it costs 80–150% more than the error does and should not be used without an estimate of the rater’s noise. Under BIAS it survives correction at 1σ and 5σ from the first rating (+7.1% [3, 11], +9.0% [5, 14]) and loses at 5σ from trial 21 (−7.7% [−12, −4]); under DRIFT only 5σ from the first rating survives (+4.0% [2, 7]), and under AR(1) nothing does.

C.2 PREDICTING THE COST FROM ONE NOISY CHOICE

Predicting algorithm behaviour from cheap landscape statistics has a long history in evolutionary computation (Weinberger, 1990; Jones & Forrest, 1995; Malan & Engelbrecht, 2013); the descriptors of Table 4 are of that kind. The predictor below is a one-trial version of the selection loss itself.

Define $\text{frag}(\sigma_e)$ as the expected loss from a single greedy choice among m quasi-random candidates when each is observed with $\mathcal{N}(0, \sigma_e^2)$ error:

$$\text{frag}(\sigma_e) = \mathbb{E}_\epsilon \left[\max_i g_i - g_{\arg \max_i (g_i + \epsilon_i)} \right], \quad \epsilon_i \sim \mathcal{N}(0, \sigma_e^2), \quad (7)$$

where $g_i = g(x_i)$ over a fixed Sobol design. It involves no surrogate, no sequence and no acquisition function, and at $m = 256$ it costs about 0.2 seconds per landscape. It is the expected opportunity cost of one noisy selection in the ranking-and-selection literature (Kim & Nelson, 2006), applied to a random design, and it is the selection term of Equation 5 for a study of one trial per design and no search.

With onset and error-process indicators in every model, $\text{frag}(\sigma_e)$ explains $R^2 = 0.52$ of the variation in excess regret across landscape-by-condition cells, against 0.04 for the onset and error-process indicators alone and 0.16 for those indicators plus the error magnitude, and nearly as much as the primary landscape descriptors together with the error magnitude ($R^2 = 0.53$; Table 20; Figure 4). Jointly they reach $R^2 = 0.70$, and in that additive model the error magnitude collapses to $\beta = -0.07$ with $p = 0.44$. That null belongs to the additive specification: with the magnitude-by- opt_z interaction Appendix B.2 found, and a quadratic in the magnitude, the descriptor model reaches $R^2 = 0.78$ with no frag term; frag still adds to it, keeping $\beta = +0.31$ [0.18, 0.45] at $R^2 = 0.79$, but the magnitude is significant again beside it (+0.17 [0.00, +0.31]; forming the product from uncentred logs instead reports this coefficient as a slope at $\text{opt}_z = 1$, which only powell is near, and flips its sign). **One noisy choice predicts, on its own, two thirds of what the best descriptor model explains, and stays informative beside it. It does not make the error magnitude redundant.** Dimension and ruggedness are not detectably different from zero here (intervals [−0.26, 0.12] and [−0.34, 0.15]), though the fixed- opt_z ladder of Appendix D.3 shows dimension does raise the cost.

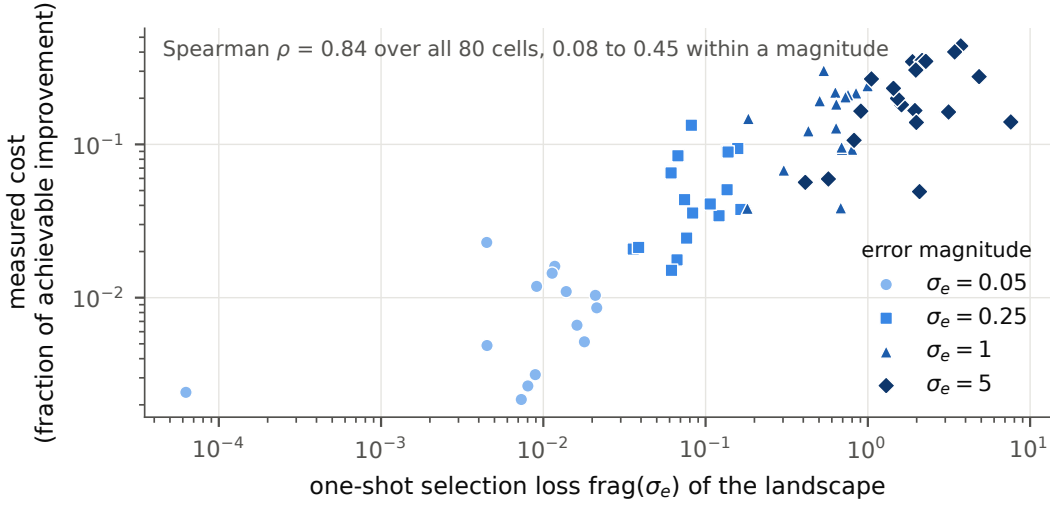


Figure 4: The one-shot selection loss $\text{frag}(\sigma_e)$ of a landscape (Equation 7, computed on 256 quasi-random designs with no optimizer) against the trajectory loss the sweep measured on it, one point per landscape and magnitude, error from the first rating. Shape and shade both encode the magnitude. Nine of the eighty cells cannot be placed on log axes, eight because frag is exactly zero and one because the measured cost is negative. Both correlations in the panel are over all eighty. The pooled value largely restates that both quantities grow with σ_e , which is why the within-magnitude range is given beside it.

Table 20: Predicting the cost from one noisy choice. Fitted on landscape $\times$ condition cell means, predictors z -scored, every model also carrying onset and error-process indicators. The interaction models add $(\log \sigma_e)^2$ as well. Intervals and p -values from a 2000-replicate bootstrap over landscapes.

model	$\text{frag}(\sigma_e)$	$\log \sigma_e$	$\log \text{opt}_z$	$\log \sigma_e \times \log \text{opt}_z$	$\log \text{tail}$	rugged.	d	R^2
indicators only	—	—	—	—	—	—	—	0.040
$\log \sigma_e$ only	—	+0.37***	—	—	—	—	—	0.158
frag only	+0.75***	—	—	—	—	—	—	0.520
descriptors only	—	+0.37***	+0.64***	—	+0.20**	-0.09	-0.03	0.526
both	+0.66***	-0.07	+0.42**	—	+0.20**	-0.06	-0.03	0.701
descriptors + int.	—	+0.37***	+0.64***	+0.54***	+0.20**	-0.09	-0.03	0.779
both + int.	+0.31***	+0.17*	+0.54**	+0.40***	+0.20**	-0.08	-0.03	0.792

Equation 7 needs the objective at a quasi-random design (here 256 points) and an estimate of rater noise; computed instead on a GP fitted to the first ten trials of a clean run, it tracks the exact value (Spearman 0.94) and predicts the measured cost as well (0.85 against 0.84; Appendix C.3). Its residuals are positive in opt_z (+0.42) and tail weight (+0.20): on tall, heavy-tailed landscapes fifty noisy choices cost more than one predicts. Out of sample, leaving one landscape out, frag with the indicators predicts at $R^2 = 0.44$ against 0.41 for the descriptors and the magnitude.

C.3 THE ONE-SHOT LOSS FROM A PILOT

Equation 7 is defined on the exact objective at 256 quasi-random designs, which a practitioner does not have. For each landscape and each of the ten seeds we fit the study’s own GP to the first k trials of a clean LogEI run and evaluate Equation 7 on its posterior mean, with the same 256 designs and the same 4,000 error draws as the exact value (Table 21).

Ten trials are enough. Pooled over magnitudes, frag from the first ten trials correlates 0.94 with the exact value and 0.85 with the measured cost, against 0.84 for the exact frag itself; twenty and fifty trials add almost nothing. Within a single magnitude the ranking of landscapes is weak for the exact frag too (0.08 to 0.45), so the predictive power is mostly across magnitudes; at 5σ the pilot

Table 21: Spearman correlations across the twenty landscapes of frag computed from the GP posterior mean after k clean trials (averaged over ten seeds) with the exact frag and with the measured cost at the early onset. Pooled rows use all eighty landscape-by-magnitude cells.

frag computed from	with exact frag	with measured cost		
	pooled	pooled	1σ	5σ
first 10 trials	0.94	0.85	0.32	0.61
first 20 trials	0.96	0.85	0.20	0.66
first 50 trials	0.96	0.85	0.15	0.63
exact objective	1.00	0.84	0.42	0.45

underestimates frag by about a third on average (a bias of -0.78) but ranks landscapes by cost better than the exact version (0.61 against 0.45).

D ABLATIONS AND THE INPUT-ERROR ARM

D.1 DOES THE SURROGATE KNOW IT IS BEING LIED TO?

The GP fits its observation noise as a free hyperparameter under a prior that shrinks it, so a measured cost of error could in principle be the surrogate’s failure to notice the error rather than the information the error destroys. This is the first thing to rule out, because if it were the explanation then every number above would be a statement about BoTorch’s default prior rather than about noisy feedback.

The mis-estimation is real, and it lands where it would do the most damage (Table 22). Refitting each surrogate from the sweep’s own logs using only its first n observations, at 1σ and above the fitted noise standard deviation is about an eighth of the injected one over the first fifteen of fifty observations, 0.56–0.66 of it by the twenty-fifth and 0.83–0.89 by the fiftieth, while at 0.05σ it is overestimated by a quarter to a half throughout, so a diagnostic evaluated only at $n = 50$ looks reassuring and is misleading. Refitting with a wide noise prior recovers the injected level far sooner (0.55 rather than 0.14 at $n = 15$, 1σ), so the shrinkage is the prior’s doing and not a limit of the data.

Table 22: Fitted over injected observation-noise standard deviation, as a function of how many observations the surrogate has seen, for error present from the first rating. The estimate is badly wrong exactly while the trajectory is being decided, and Table 23 shows that this costs almost nothing.

observations	0.05σ	0.25σ	1σ	5σ
8	1.43	0.30	0.12	0.11
15	1.50	0.41	0.14	0.10
25	1.37	0.72	0.66	0.56
50	1.24	0.88	0.83	0.89

It was natural to expect this to explain the onset effect of Section 4: from the first rating the error arrives in exactly the window where the noise cannot yet be estimated, whereas at a mid-run onset the surrogate has a settled lengthscale and thirty trials in which to learn. **We tested that directly and it is wrong.** Supplying the true injected variance as `train_Yvar` and re-running 4,800 paired cells removes about 3% of the summed cost of the eight cells in Table 23 (+15% at 0.25σ from the first rating, -1% and $+4\%$ at 1σ and 5σ , +18%, +1% and +2% from trial 21; the 0.05σ cells have reference costs of 0.006 and 0.001 and their shares are not meaningful), no condition survives false-discovery-rate correction, and the onset ratio at 1σ does not shrink: it is 5.1 before and 5.1 after. Per acquisition the effect is small and inconsistent in sign, removing 6–10% of the cost for PI, EI, qNEI and UCB, 0% for LogEI, and *adding* 18% for qUCB.

Two conclusions follow. Bayesian optimization is insensitive to its own noise hyperparameter in this regime, despite getting it badly wrong for the first fifteen trials, worth knowing on its own, and a caution against reading too much into noise-estimation diagnostics. And the cost this paper measures is the information cost of error, not an artefact of a misspecified surrogate, which is what licenses reading Table 1 as a property of the landscape. The trajectory onset effect is the clean run’s post-onset improvement, the bound of Appendix B.3, not the surrogate’s noise estimate.

Table 23: Telling the surrogate the truth about its noise. Paired within (landscape, acquisition, magnitude, onset, seed); positive Δ means the treatment suffers more. Nothing survives false-discovery-rate correction.

σ_e	t_0	reference	treatment	Δ	share	d_z	p_{FDR}
0.05	0	0.006	0.006	-0.000	+5%	-0.02	0.992
0.25	0	0.038	0.032	-0.006	+15%	-0.18	0.992
1	0	0.117	0.118	+0.001	-1%	0.03	1.000
5	0	0.213	0.204	-0.009	+4%	-0.23	0.992
0.05	20	0.001	0.003	+0.001	-105%	0.12	0.992
0.25	20	0.010	0.008	-0.002	+18%	-0.16	0.992
1	20	0.023	0.023	-0.000	+1%	-0.02	0.992
5	20	0.047	0.046	-0.001	+2%	-0.03	1.000

D.2 IS IT THE ACQUISITION OR THE INCUMBENT?

best_f is the maximum posterior mean at visited points, which shrinks under noise and so makes improvement-based acquisitions automatically more exploratory. The acquisitions differ in their exposure to this: LogEI, EI, PI, LogPI and qEI consume best_f ; UCB and qNEI do not. A ranking of acquisitions by robustness is therefore partly a ranking of their sensitivity to a design choice that is not part of the acquisition at all. Re-running with the observed maximum as the incumbent removes 20.0% of the excess-weighted cost overall (24% at 1σ from the first rating and 20% from trial 21, and about a half to four fifths at 0.25σ and below), with six of eight conditions significant after false-discovery-rate correction (Table 24).

The per-acquisition breakdown is in Table 25. UCB and qNEI move by *exactly* zero, as they must, a control that validates the pairing at machine precision. The probability-of-improvement family moves most: PI loses 45% of its measured cost and LogPI 40%, against 12% for qEI, 9% for EI and 4% for LogEI. So a substantial part of “PI is fragile under noise” is “the posterior-mean incumbent is fragile under noise, and PI is the acquisition most exposed to it”. Every acquisition-robustness ranking here, including Table 2, is a ranking of acquisition-and-incumbent pairs, and the same is true of the applied results it is modelled on. Reporting the incumbent definition alongside the acquisition should be standard. Scored against the standard process’s clean run rather than its own, the switch still recovers a fifth of the trajectory loss, but the design it deploys is worse (Appendix E.5).

Table 24: Observed-maximum against posterior-mean incumbent, same pairing as Table 23.

σ_e	t_0	reference	treatment	Δ	share	d_z	p_{FDR}
0.05	0	0.010	0.003	-0.007	+72%	-0.56	0.019
0.25	0	0.050	0.026	-0.024	+48%	-1.39	<0.001
1	0	0.132	0.100	-0.032	+24%	-0.93	0.004
5	0	0.223	0.200	-0.022	+10%	-0.71	0.011
0.05	20	0.003	0.001	-0.002	+81%	-0.24	0.142
0.25	20	0.014	0.005	-0.009	+64%	-0.88	0.002
1	20	0.027	0.021	-0.005	+20%	-0.50	0.032
5	20	0.049	0.048	-0.000	+0%	-0.01	0.421

Table 25: The incumbent arm by acquisition. UCB and qNEI never read best_f and move by exactly zero, which validates the pairing. The probability-of-improvement family moves most.

acquisition	reference	treatment	Δ	share removed
PI	0.089	0.049	-0.040	+45%
LogPI	0.088	0.053	-0.035	+40%
qEI	0.054	0.048	-0.006	+12%
EI	0.058	0.053	-0.005	+9%
LogEI	0.056	0.054	-0.002	+4%
qNEI	0.049	0.049	+0.000	+0%
UCB	0.049	0.049	+0.000	+0%

D.3 BREAKING THE DESCRIPTOR CONFOUND

In our suite opt_z , sparsity and skew correlate at about 0.9, and for a bounded function with an isolated optimum that is close to a structural identity, so Appendix C.2’s descriptor terms cannot be separated observationally, however many real landscapes are added. Two purpose-built families vary them by construction: a cos-field background plus one spike, whose amplitude moves opt_z and whose width moves sparsity; and the Levy function at $d \in \{4, 7, 11\}$, one landscape family across a dimension ladder (Table 26).

opt_z on its own is not causal. Holding the spike narrow and raising its amplitude by $4\times$ moves opt_z from 5.0 to 21.7 and changes the excess regret at 1σ from 0.246 to 0.245, a paired difference of $-0.001 [-0.035, +0.036]$ (seed bootstrap, five seeds). At the other magnitudes the change is not zero but never significantly upward: $+0.007 [-0.012, +0.018]$ at 0.05σ , $-0.127 [-0.358, +0.004]$ at 0.25σ and $-0.062 [-0.125, -0.012]$ at 5σ . Widen the spike and the same amplitude change raises excess from 0.495 to 1.888 at 1σ ($+1.39 [+0.26, +2.53]$), and at every magnitude from 0.25σ . What the observational $\beta_{\text{opt}_z} = +0.42$ picks up is therefore an interaction, not a main effect: a taller optimum costs more to miss only when it is broad enough that the optimizer would otherwise have found it. On a true needle, the optimizer was never going to find the optimum with or without noise, so there is nothing for the noise to destroy. It also means that the opt_z exponent of Appendix B.2 describes the suite rather than a mechanism: it is not opt_z as such that makes an error expensive.

Dimension, and a within-family ladder that corrects itself. Within the Levy family, excess regret at 1σ rises monotonically from 0.114 at $d = 4$ to 0.222 at $d = 7$ and 0.264 at $d = 11$, while the cross-suite regression puts β_d at -0.03 and not significant. Read on its own that looks like a real effect the observational analysis missed. It is not, or not mostly: opt_z also rises along the Levy ladder, 1.53 to 2.07 to 2.62, so that comparison varies dimension and signal strength together.

The bump ladder separates them. Amplitude and width are chosen at each dimension so that opt_z is 9.0 and the spike occupies a 10^{-6} fraction of the box at all three, leaving dimension as the only difference. In the table’s units, excess regret is 0.28 at $d = 4$, 0.25 at $d = 7$ and 0.43 at $d = 11$. The $d = 4$ rung has headroom 0.098, just below the 0.10 screen applied everywhere else, so the comparison that clears the screen is $d = 7$ to 11: there the fixed- opt_z ladder rises $1.7\times [1.1, 5.3]$ at 1σ and $1.7\times [1.4, 2.3]$ at 5σ (ten seeds, paired), where Levy rises $1.2\times [0.6, 1.9]$; at 0.25σ the interval, $[0.6, 3.7]$, includes no change, and at 1σ the excluded $d = 4$ rung lies above $d = 7$. **With opt_z held fixed, dimension raises the cost at 1σ and 5σ on a two-point comparison. What the Levy ladder cannot say is how much of its $2.3\times$ is dimension rather than signal strength.** What rises monotonically with dimension on both ladders is the inference metric, the true value of the design a practitioner would deploy: 0.52, 0.70, 0.99 on the fixed- opt_z ladder. Higher dimension makes the progress harder to identify from noisy observations at the end as well as harder to make.

The methodological lesson is that a within-family ladder is not automatically a manipulation of the property it is indexed by: the Levy ladder varied signal strength along with dimension, and ours became a manipulation of dimension only when it was built to hold opt_z fixed.

Table 26: Purpose-built landscapes. Raw post-onset excess regret in landscape SDs, GAUSSIAN error from the first rating. Amplitude and width are varied independently in the top block. The middle block is one function family at three dimensions. The bottom block holds $\text{opt}_z = 9.0$ and the spike volume fixed across three dimensions.

landscape	d	opt_z	sparsity	0.05σ	0.25σ	1σ	5σ
bump_a4_w0.05	4	5.0	$<1.5\text{e-}05$	0.007	0.152	0.246	0.324
bump_a16_w0.05	4	21.7	$<1.5\text{e-}05$	0.014	0.026	0.245	0.262
bump_a4_w0.15	4	5.3	$3.8\text{e-}04$	0.133	0.289	0.495	0.704
bump_a16_w0.15	4	12.3	$1.8\text{e-}04$	0.298	0.730	1.888	3.173
levy_4d	4	1.5	$2.6\text{e-}01$	0.006	0.048	0.114	0.200
levy_7d	7	2.1	$7.3\text{e-}02$	0.051	0.115	0.222	0.305
levy_10	11	2.6	$1.3\text{e-}02$	0.001	0.147	0.264	0.481
bump_d4	4	9.0	$<1.5\text{e-}05$	0.001	0.121	0.279	0.327
bump_d7	7	9.0	$<1.5\text{e-}05$	-0.001	0.103	0.248	0.428
bump_d11	11	9.0	$<1.5\text{e-}05$	0.142	0.215	0.427	0.734

D.4 THE INPUT-ERROR ARM

Every error process above corrupts the reported value. People also make the other mistake: a slider overshoots, the wrong option is pressed, a configuration is applied to the wrong condition. The optimizer proposes x , the person experiences $x' \neq x$, rates it honestly, and the rating is filed against x : the right value at the wrong location. A fitted oracle cannot score it, since that needs the objective at x' .

A SLIP displaces every trial by $\delta \sim \mathcal{N}(0, s^2)$ per coordinate, with s a fraction of that coordinate’s range, and clips the result to the box. A MISCLICK leaves the trial exact with probability $1 - p$ and otherwise lands on a uniformly random design. Both magnitudes are swept over $\{0.01, 0.05, 0.15, 0.4\}$ at the same onsets and budget, with six model-based acquisitions (LogEI, EI, PI, UCB, qUCB, qNEI), the two floors and five seeds. Neither is in landscape SDs, so the comparison with the response-error grid is on the damage axis only.

If the slip goes unnoticed the surrogate trains on $(x, g(x'))$ and the deployed design is the recorded x , so we score it there. A counterfactual arm records $(x', g(x'))$, as an instrumented interface would, with identical slip draws. The contrast separates the cost of evaluating the wrong point from that of mislabelling it. The model-free floors are not a zero control here: they evaluate the wrong points without learning, so their excess is the geometric cost of a slip alone.

A slip costs about as much as a rating error. A slip of 5% of each coordinate’s range on every trial, from the first, destroys 5.4% of the achievable improvement; one of 15% destroys 11.5%; one of 40%, 23.2% (Table 27). On this arm’s six acquisitions and seeds under GAUSSIAN error, the 0.25σ , 1σ and 5σ rating errors of Table 1 cost 3.8%, 11.7% and 21.3% (the $T = 50$ row of Table 15; pooled: 5.3%, 14.1%, 22.0%): a 5% slip costs more than a 0.25σ rating error, a 15% slip about as much as 1σ , a 40% slip about as much as 5σ . A misclick is cheaper at every matched magnitude: even when 40% of trials land somewhere random it costs 13.5%, little more than a 15% slip on every trial and a little over half of a 40% one. A late slip is cheaper than an early one by less than a late rating error: $1.4\text{--}3.6\times$ against $5.1\times$ at 1σ on the same basis.

Half of a small slip’s cost is the record, not the displacement. The decomposition in Table 27 (in full, Table 28) separates the mechanisms. The model-free floor barely moves, since a random design displaced is still random; only at 40%, where clipping piles evaluations at the box edges, does it lose 6.4% [0, 13]. A learner with a correct record loses far more, 2.8% at a 5% slip and 19.6% at 40%: a displaced proposal is a worse one. Mislabelling adds 2.7 points at 5% and 4.4 at 15%, half and two-fifths of the total (paired intervals [0.35, 0.61] and [0.26, 0.49]; not estimable at 1%); at 40% the displacement dominates and the mislabelled share falls to a sixth.

The design in the log is not the design that earned its rating. Scored at the recorded x a practitioner would deploy, the final design’s excess regret is 16.4% of the achievable improvement at

Table 27: Fraction of the achievable improvement destroyed by an input error. A slip’s magnitude is the displacement SD as a fraction of each coordinate’s range; a misclick’s is the probability of a random trial. The last three columns decompose the early-onset slip: *floor*, the model-free arms; *logged*, a learner whose record holds the point evaluated; *mislabelling*, the paired difference between the unnoticed slip (second column) and *logged*. Brackets: 95% intervals over landscapes.

magnitude	SLIP		MISCLICK		SLIP from trial 1, decomposed		
	from trial 1	from trial 21	from trial 1	from trial 21	floor	logged	mislabelling
1%	1.5 [0, 3]	1.1 [0, 2]	0.4 [0, 1]	0.1 [0, 0]	0.3 [0, 1]	0.7 [0, 2]	0.8 [0, 2]
5%	5.4 [3, 8]	3.0 [2, 5]	2.8 [2, 4]	0.5 [0, 1]	0.2 [−1, 1]	2.8 [2, 4]	2.7 [1, 4]
15%	11.5 [8, 15]	5.0 [3, 8]	6.8 [5, 9]	2.0 [1, 3]	1.2 [−1, 3]	7.1 [5, 10]	4.4 [2, 6]
40%	23.2 [16, 31]	6.4 [4, 10]	13.5 [10, 18]	3.5 [2, 6]	6.4 [0, 13]	19.6 [12, 28]	3.6 [2, 6]

a 5% slip from the first trial, 34.3% at 15% and 65.8% at 40%, against 6.5%, 13.4% and 24.9% for the best design the person experienced (Appendix D.5): a factor of 2.5 to 2.6, and 1.6 at a 1% slip. The optimizer’s progress survives an input error far better than the record of it, which is what a deployment inherits.

D.5 INPUT ERROR: THE DESIGN THAT SHIPS

Under an unnoticed slip the optimizer’s record holds the proposal x , not the point x' it evaluated, and the design a practitioner deploys is the recorded one. Table 29 scores both at the end of the run: the best design the person experienced, and the design the log recommends, scored where the log says it is.

Table 28: Where a slip’s cost comes from, at both onsets. *floor*: the model-free arms, which evaluate displaced points but learn nothing; *logged*: a learner whose record holds the point evaluated; *unnoticed*: the same learner with the record holding the proposal; *mislabelling*: the paired difference of the last two, on shared seeds and slip draws. The early-onset rows also appear in Table 27.

	SLIP	floor	logged	unnoticed	mislabelling
<i>error from trial 1</i>					
1%		0.3 [0, 1]	0.7 [0, 2]	1.5 [0, 3]	0.8 [0, 2]
5%		0.2 [−1, 1]	2.8 [2, 4]	5.4 [3, 8]	2.7 [1, 4]
15%		1.2 [−1, 3]	7.1 [5, 10]	11.5 [8, 15]	4.4 [2, 6]
40%		6.4 [0, 13]	19.6 [12, 28]	23.2 [16, 31]	3.6 [2, 6]
<i>error from trial 21</i>					
1%		0.0 [0, 0]	0.2 [0, 1]	1.1 [0, 2]	0.9 [0, 1]
5%		−0.4 [−1, 0]	2.0 [1, 3]	3.0 [2, 5]	1.0 [0, 2]
15%		−0.4 [−1, 1]	4.5 [2, 7]	5.0 [3, 8]	0.5 [0, 1]
40%		−0.6 [−2, 1]	6.3 [3, 10]	6.4 [4, 10]	0.1 [0, 0]

Table 29: Final-trial excess regret under an unnoticed slip, as a percentage of the achievable improvement, over the model-based acquisitions. *Evaluated*: the best true value among the points touched. *Deployed*: the true value at the recorded location of the best-rated trial. Brackets are 95% cluster-bootstrap intervals over landscapes.

SLIP	from trial 1		from trial 21	
	evaluated	deployed	evaluated	deployed
1%	1.7 [0, 3]	2.7 [1, 5]	1.5 [1, 3]	2.4 [1, 4]
5%	6.5 [4, 10]	16.4 [9, 24]	3.9 [2, 6]	9.6 [5, 15]
15%	13.4 [8, 20]	34.3 [24, 45]	6.9 [3, 10]	13.4 [8, 19]
40%	24.9 [16, 34]	65.8 [53, 78]	9.0 [5, 14]	13.3 [8, 19]

E REMEDIES

E.1 END-OF-STUDY PROCEDURES

Both procedures keep T fixed and spend the final trials differently, so each is replayed exactly from an existing log: truncate at the trial the procedure starts, refit the loop’s GP on that prefix, and simulate only the new trials. The premise is checked against the re-rating arm, whose earlier trials must equal the standard run’s.

Sitting(k), the tournament of the code. Search trials $1..T - k$. The last k are one sitting in which each of k candidates is shown once. Candidates are the top k visited designs by the posterior mean less one latent SD, or by observed rating. The winner is the best look, or the posterior mean after conditioning on the looks. Error shared by every trial of a sitting cancels under comparison, so only the idiosyncratic part enters: σ for GAUSSIAN, BIAS and DRIFT, the innovation for AR(1). That is the model everywhere in the paper. The replay also ran the looks at half that SD, a sensitivity for which we can name no mechanism: a comparison between two looks carries the noise of both, $\sqrt{2}$ times one rating’s (Appendix E.7), not half of it. At half the SD the sitting over the top five recovers 19.7% [13, 27] of GAUSSIAN error’s deployed loss rather than 12.2%, and the fixed $k = 16$ gains +0.051 rather than +0.022. Every headline figure in the paper is at the full SD. At the full SD the sitting over the top five recovers 26.3% of the deployed loss at 5σ from trial 21 and 23.8% under an unnoticed slip.

Confirmation(k). Search trials $1..T - 2k$. The last $2k$ rate the chosen design and the best initial design k times each in counterbalanced order. The chosen design ships if its *mean* rating over those k is at least the initial design’s, and the study *claims* an improvement only if a one-sided paired t -test at $\alpha = 0.05$ prefers it. The two decisions are separate, which is why this changes little about the design that ships (it recovers 3.2%–3.4% of the cost) and a great deal about what the study may claim. The standard process claims an improvement in 98.3% of runs and that claim is false in 15.7% of them, rising to 42.7% at 5σ from the first rating. Four confirmation trials take the false-claim rate to 0.5% overall and 1.0% at 5σ from the first rating, at the cost of detecting only 50.7% of the real improvements (25.9% at 1σ from the first rating); eight trials give 0.46% and 65.7%. Two changes are bundled there, and they are worth separating, because the standard process applies no test at all: it claims whenever the best observed rating of the run exceeds the best of the initial design, which is true unless the two coincide, so its power is 1.000 and its size 0.75–1.00 in the GAUSSIAN cells. Claiming on the fresh mean alone, with the trials spent but no test, already takes the false-claim rate from 15.7% to 6.6%. The $\alpha = 0.05$ test takes it from 6.6% to 0.5%. Of the 15.2-point reduction the trials therefore buy 9.1 points, 60%, and the test 6.1. The test is not free: claiming on the fresh mean detects 88% of the real improvements, and adding the test cuts that to 51%. Power is also strongly magnitude-dependent: 87.5% at 0.05σ and 6.8% at 5σ from the first rating, so the pooled 50.7% is carried by the cells where little is at stake. All figures in this paragraph are the GAUSSIAN cells.

E.2 DERIVING THE SPLIT FROM THE RUN IN HAND

Write μ_i, σ_i for the loop GP’s posterior latent mean and SD at the distinct designs visited by the decision point, ordered by the posterior mean less one latent SD, and s for the idiosyncratic rating SD of a sitting. A sitting of k shows the top k candidates once each, so with $f_i \sim \mathcal{N}(\mu_i, \sigma_i^2)$ and looks $y_i = f_i + \mathcal{N}(0, s^2)$ the identification term is $U(k) = \mathbb{E}[f_{\arg \max_i y_i}]$, evaluated by Monte Carlo. It is not monotone in k : a larger candidate set is more likely to contain the best visited design and more likely to let a worse one win on a lucky look. The search term prices the forgone trials by the run’s own gain in best rating per trial over the last ten, floored at zero. Every policy (ship by posterior mean, ship by the posterior mean less one latent SD, or a sitting of k) is scored under the single posterior at $n_0 = T - \max k$, the last trial at which all of them are still open; scoring each k under its own $T - k$ posterior looks more faithful and is not, because two GPs fitted on different numbers of points put their maxima on different scales, and the comparison then picks up that difference rather than the difference between policies. The sitting’s SD is read from the surrogate’s fitted noise, which is the noise of the *standardised* targets, and multiplied by the outcome transform’s scale so that it and the posterior sit on one axis; without that conversion a sitting would appear up to 5.6 times more precise than it is. The rule reads only ratings and the surrogate. The

true objective enters only when the choice is scored, against the realised regret of that (run, k) in the replay.

E.3 HELD-OUT VALIDATION OF THE SELECTED PROCEDURES

Section 7 reports procedures chosen from families of replayed variants: nine values of k , nineteen sitting variants, four ship rules and two rank rules. The best of a family scored on the data it was chosen on is biased upward, so every such choice is re-made here on seeds 7–11 and scored on seeds 12–16, with the reverse direction as a check, and separately on 1000 random splits of the landscapes into ten and ten, each used both ways. The replays of the main sweep cover seeds 7–16, so this needs no new simulation; the gross-fault and capped-scale arms were extended to seeds 12–16 for it.

Table 30: Each selection re-made on training data and scored on data it never saw. Gains are in units of the achievable improvement; recoveries are shares of GAUSSIAN error’s deployed loss (the ship rule at 0.25σ and above; the rank rules on their own arms). The landscape column is the median and 95% range over 2000 folds of the held-out value of the choice made on the other ten landscapes. ^aSeeds 7–11, on which the rank rules were chosen; their landscape splits use the same seeds.

family	chosen on 7–11	all seeds	seeds 12–16	landscape splits
k of the sitting	$k = 12$	+0.021	+0.017 [+0.001, +0.033]	+0.019 [+0.000, +0.034]
sitting, 11 at full SD	top 3, look	13.2%	16.1% [9, 23]	12.2% [2, 19]
top five, fixed	top 5, look	12.2%	17.4% [9, 26]	—
ship rule	LCB, 2 SD	10.1%	10.7% [5, 16]	10.1% [4, 16]
rank rule, gross faults	LCB on ranks	50.5% ^a	48.8% [34, 63]	50.0% [35, 63]
rank rule, capped scale	LCB on ranks	36.1% ^a	37.9% [31, 45]	35.6% [28, 44]

The optimism of each choice is small. Net of a shift between the seed sets that moves every candidate alike (the test seeds score about seven points higher for every sitting variant, which is why several test values exceed the full-data ones), it is +0.003 to +0.006 of the achievable improvement for k and -0.2 to $+3.5$ points of recovery for the sitting variants, and the ship rule is chosen identically on every split, the rank rule on 66 to 90% of them. The chosen k is between 5 and 16 on all but 0.1% of splits, so no single k (12 chosen on seeds 7–11, 8 on seeds 12–16) is a stable point estimate, while “ k between 5 and 16 gains about 0.02” holds out of sample.

Multiplicity. On the test seeds we tested each procedure’s twenty per-landscape gains with a two-sided Wilcoxon signed-rank test and applied Holm’s correction over the 27 procedures of the core family (sitting variants, ship rules, shortlist sizes, rank rules) and over all 41 replayed procedures. With twenty landscapes the smallest attainable p is 1.9×10^{-6} and Holm’s first threshold $0.05/27$, so a procedure must gain on nearly every landscape to survive. The shortlist of two, three or five designs survives (it gains on 18, 20 and 20 landscapes), as do five sitting variants at half the idiosyncratic SD, the sensitivity of Appendix E.1. No pooled sitting at the full SD survives the 27-procedure family (the top-five sitting gains on 15 of 20 landscapes, adjusted $p = 0.09$), nor any ship rule (the cautious rule, 16 of 20, $p = 0.06$), fixed k as a gain ($k = 8$, 13 of 20, $p = 0.27$) or rank rule on the main sweep; $k = 30$ survives as a significant loss. On the test seeds the derived budget rule gains -0.007 [-0.022 , $+0.006$], 0.024 [0.009 , 0.039] below the k chosen on seeds 7–11.

The sitting cell by cell. The pooled k -curve mixes magnitudes at which the sitting pays with magnitudes at which it costs. Table 31 re-makes the choice of k inside each magnitude and on-set (`scripts/sitting_by_magnitude.py`, LogEI and qNEI, the four processes, looks at the full idiosyncratic SD). At 0.05σ and 0.25σ every size loses, since the sitting’s price exceeds what it can find. At 1σ from the first rating both seed directions choose $k = 2$. It gains $+0.030$ [$+0.012$, $+0.050$] on seeds 12–16 and $+0.021$ [$+0.011$, $+0.031$] on seeds 7–11, and recovers 12% [5, 18] of the cost of error on seeds 12–16. It is positive under each process ($+0.012$ for AR(1) to $+0.047$ for BIAS) and for both acquisitions, and 2,000 landscape-split folds choose it 88% of the time (median held-out gain $+0.025$ [$+0.009$, $+0.038$]). A two-way bootstrap that also resamples the five held-out seeds, which every landscape shares, gives [$+0.010$, $+0.053$]. Larger sittings pay at 5σ ($k = 8$ from the first rating, $k = 16$ from trial 21) and at 1σ from trial 21, where every size from 3 to 12 gains about 0.02 on all seeds. The pooled gain of $k = 12$ is carried by the 5σ cell from trial 21. Without it that gain is 0.000 [-0.012 , $+0.012$] on all seeds and -0.002 [-0.016 , $+0.011$]

on seeds 12–16, and the two-way interval of the pooled $k = 12$ on seeds 12–16 is $[-0.003, +0.039]$. This analysis followed the pooled one. The cell the main text reports is the one the anchor of Section 4 fixes, and Holm’s correction is over the nine sizes within it. Over all eight cells and nine sizes, Holm’s adjusted p for it is 0.23. In the reverse seed direction the top-five sitting of Table 30 (GAUSSIAN error, LogEI, qNEI and UCB) recovers 6.5% [0, 13] of GAUSSIAN error’s cost on seeds 7–11, against 17.4% on seeds 12–16. Under misclicks, which cost little (0.068), it recovers -24.5% $[-45, -10]$, since its fixed price of 0.024 outweighs what it finds.

Table 31: The final sitting cell by cell. k is chosen on seeds 7–11 and its gain scored on seeds 12–16, in units of the achievable improvement, with a landscape-bootstrap interval. Landscapes counts those that gain. Holm p is the adjusted p of a two-sided Wilcoxon signed-rank test over the twenty held-out per-landscape gains, corrected over the nine values of k within the cell. The last two columns give the k chosen in the reverse direction and the gain of $k = 12$, the pooled choice, on all seeds. LogEI and qNEI, the four error processes, looks at the full idiosyncratic SD.

cell	k , 7–11	gain on seeds 12–16	landscapes	Holm p	k , 12–16	$k = 12$
pooled	12	+0.017 [+0.001, +0.033]	14	0.29	8	+0.021
0.05 σ , from trial 1	3	-0.010 [-0.020, -0.003]	7	0.12	5	-0.018
0.05 σ , from trial 21	5	-0.014 [-0.024, -0.005]	6	0.06	3	-0.020
0.25 σ , from trial 1	5	-0.006 [-0.014, +0.003]	10	1.00	3	-0.017
0.25 σ , from trial 21	5	-0.006 [-0.018, +0.004]	11	1.00	3	-0.011
1 σ , from trial 1	2	+0.030 [+0.012, +0.050]	14	0.03	2	-0.010
1 σ , from trial 21	16	+0.011 [-0.012, +0.033]	13	0.88	5	+0.019
5 σ , from trial 1	8	+0.058 [+0.017, +0.102]	17	0.03	5	+0.059
5 σ , from trial 21	16	+0.227 [+0.147, +0.321]	19	< 0.001	20	+0.163

E.4 BUDGET-NEUTRAL ARMS

Every arm below keeps the number of human trials equal to the standard process or lowers it, and is scored against the standard process by the estimand of Appendix E.5: twenty landscapes, seeds 7–11, both onsets where the arm has them. Table 33 gives the share of the standard process’s cost of error that each recovers on the design the study deploys.

The faults in the table are defined as follows. A *spike* adds, with probability 0.05 or 0.15 per trial, an independent draw of 5 or 20 landscape SDs on top of the rating error. The clipping remedy truncates every rating to the landscape’s sampled range, the analogue of a bounded rating scale, and the relevance-pursuit remedy (Ament et al., 2024) lets a sparse set of trials carry their own inflated noise variance. A *relay* of raters gives each rater a fixed offset, drawn with twice the SD of the rating error, and hands over either every ten trials (block) or trial by trial among five raters (round robin). The remedy estimates the offsets inside the GP fit. A *rating cap* truncates every noisy rating at the landscape’s 0.9 quantile, so the best designs become indistinguishable. The re-anchored remedy lifts the cap to half a landscape SD above the true value of the best design rated so far. *Halo* error, in the multi-objective arm, makes one trial’s rating errors share a common factor across objectives, at correlation 0.85. The backfit estimates that factor from the residuals and refits. A *lost rating* never arrives, either at random or only for designs below the landscape’s 40th percentile. The standard process drops the trial, and the arm instead imputes the 20th percentile of the ratings so far and never deploys the imputed design.

The pattern is not that everything fails. It is that changes to the search fail, and what works is an explicit model of how the rating was produced. Clipping the response to the observed range recovers 6–35% of the cost of gross faults at a price of exactly zero; a relevance-pursuit likelihood recovers 26–38% when 15% of trials carry a gross fault and loses when only 5% do; fitting per-rater offsets recovers 32% [23, 40] when raters alternate and nothing when they hand over in blocks, because a block offset is confounded with the search’s own time trend while an interleaved one is identifiable; re-anchoring a saturating cap recovers 25% [17, 32]; and removing the factor shared across objectives recovers 25% [7, 50] of the deployed loss of halo error. A saturating rating scale is itself among the most expensive faults we measured, costing 185% [132, 258] more than the same rating noise on a scale that does not saturate.

Table 32 splits each search-side arm’s change in the deployed regret, and the self-report arm’s, into its search and selection parts. A change that lowered selection loss and raised search loss by more would appear above as a null; one arm does this, the lower-confidence-bound incumbent, and its net effect is a clear loss rather than a null. Its selection gain is the same with LogEI and with LogPI (−2.2 points each, only the second interval excluding zero), and its net loss is the search loss of its LogPI half. Most arms raise selection loss, and the minimum-distance constraint finds better designs than the standard process while shipping worse ones. The self-reported confidence, an observation-model arm listed for contrast, is the one arm that lowers both terms, the search term within noise (−1.2 [−2.2, +0.1]).

Table 32: Where each arm’s change in the deployed regret comes from: the difference from the like-for-like standard process at the final trial, split by Equation 5, in points of the achievable improvement (positive is worse), pooled over the arm’s magnitudes and onsets, with the deployed difference’s landscape-bootstrap interval. Price is the deployed difference without error, which falls entirely on search, since a clean run has no selection loss.

arm	Δ deployed	Δ search	Δ selection	price
Augmented expected improvement	+1.1 [−2.1, +3.9]	+0.1	+1.1	+1.4
Thompson sampling	+13.4 [+9.1, +18.4]	+5.7	+7.7	+8.1
Lower-confidence-bound incumbent	+5.4 [+2.4, +8.2]	+7.6	−2.2	+2.8
Minimum-distance proposals	+3.7 [+1.4, +5.8]	−1.9	+5.7	+4.0
Slip-aware acquisition, 0.05 slip	+0.5 [−0.5, +1.5]	+0.5	+0.0	+2.8
Slip-aware acquisition, 0.15 slip	+4.4 [+1.8, +7.2]	+2.0	+2.4	+7.3
Slip-aware acquisition, 0.4 slip	+13.7 [+5.7, +22.6]	+6.1	+7.6	+14.6
Effort front-loaded to the first ten trials	+3.2 [−1.8, +9.6]	−1.4	+4.6	0
Effort front-loaded to the first twenty	+4.1 [−2.7, +13.6]	−1.0	+5.0	0
Effort split between first and last trials	+10.3 [+5.3, +15.4]	−0.2	+10.4	0
Effort on the last ten trials	+5.7 [+1.0, +11.0]	+1.1	+4.6	0
Self-reported confidence as a variance	−3.2 [−5.3, −1.3]	−1.2	−2.0	−0.1

E.5 PROCESS ADAPTATIONS

The measurements above say what error costs a fixed process. The practical question is whether the process can be changed to cost less, and the scoring has to change with it. Rating a design twice, or re-rating designs before deploying, also changes the clean run, because a repeated rating of an exact objective carries no information, so a process change scored by its excess over its own clean run is measured against a handicapped reference. We score each change against the *standard* process instead. Per paired cell (landscape, acquisition, magnitude, onset, seed), with regret as a fraction of the achievable improvement,

$$\text{recovered} = \frac{R_{\text{noisy}}^{\text{std}} - R_{\text{noisy}}^{\text{adapt}}}{R_{\text{noisy}}^{\text{std}} - R_{\text{clean}}^{\text{std}}}, \quad \text{price} = R_{\text{clean}}^{\text{adapt}} - R_{\text{clean}}^{\text{std}},$$

aggregated as ratios of landscape means with landscape-bootstrap intervals, on two responses: the post-onset trajectory regret and the regret of the design the experimenter would deploy. qKG, re-rating the incumbent, augmented EI and Thompson sampling replace the acquisition, so there the standard process is LogEI, the suite’s default. Every arm runs twenty landscapes, seeds 7–11, both onsets and the error process named in Table 34, with LogEI and qNEI unless stated. The incumbent and known-noise rows re-score the arms of Appendix D.1 and D.2 with their own acquisitions.

Scoring an arm that replaces the acquisition against the mean of the ten standard acquisitions is not like for like: that reference carries the four weakest acquisitions’ cost, and augmented EI then “recovers” +373% on the trajectory at 0.05σ from trial 21, which would mean the arm under error beats the standard process with no error at all. The ten-mean figures are kept beside the LogEI ones in the analysis output. Shares of a cost are ratios of landscape means in units of the achievable improvement. Formed from raw landscape-SD values instead, the landscapes with the largest opt_z dominate, and the incumbent arm’s overall figure reads 15.6% rather than 20.0%.

Two of the paper’s own ablations, re-scored. The observed-max incumbent and the known-noise GP change the clean run too, so their shares in Appendix D.1 and D.2 are also excess over their own

Table 33: Budget-neutral arms: share of the standard process’s cost of error recovered on the deployed design, with 95% landscape-bootstrap intervals. Negative means the change does worse under error than the standard process. Each arm is scored on the magnitudes and onsets it was run at, which differ between arms, so a row is a share of its own reference’s cost and the column is not a ranking. The gross-fault rows are 0.25σ from the first rating only. The lost-rating arm divides by a reference cost of 0.010, near zero and negative on nine of twenty landscapes, so it is given as an absolute loss in points of achievable improvement rather than a share. Above the rule, changes to the search; below it, changes to the model of how the rating was produced.

Arm	Recovered
Augmented expected improvement	−5% [−21, +10]
Slip-aware acquisition, 0.05 slip	−4% [−15, +4]
Effort front-loaded to the first ten trials	−7% [−20, +4]
Effort front-loaded to the first twenty	−8% [−31, +5]
Effort on the last ten trials	−12% [−24, −2]
Minimum-distance proposals	−13% [−23, −6]
Slip-aware acquisition, 0.15 slip	−18% [−34, −9]
Effort split between first and last trials	−21% [−33, −10]
Lower-confidence-bound incumbent	−27% [−42, −11]
Slip-aware acquisition, 0.4 slip	−35% [−54, −16]
Thompson sampling	−65% [−91, −42]
Relevance pursuit, 15% of trials spiking at 20 SD	+38% [+24, +51]
Response clipped to range, 5% spiking at 20 SD	+35% [+23, +46]
Per-rater offsets, round-robin handover	+32% [+23, +40]
Relevance pursuit, 15% spiking at 5 SD	+26% [+7, +42]
Re-anchored rating cap	+25% [+17, +32]
Halo backfit across objectives	+25% [+7, +50]
Response clipped to range, 5% spiking at 5 SD	+22% [+10, +35]
Response clipped to range, 15% spiking at 20 SD	+22% [+4, +38]
Response clipped to range, 15% spiking at 5 SD	+6% [−7, +18]
Per-rater offsets, block handover	+5% [−5, +14]
A rating lost at random, imputed low rather than dropped	−9.4 points
Relevance pursuit, 5% spiking at 20 SD	−30% [−119, +12]

Table 34: Share of the standard process’s cost of error that each process change recovers, pooled over magnitudes and onsets, with 95% landscape-bootstrap intervals (a star when the Benjamini–Hochberg-corrected Wilcoxon p is below 0.05), and the price the change pays without error in percentage points of the achievable improvement. Negative recovery means the adapted process does worse under error than the standard one. The fitted-oracle datasets have no achievable-improvement scale.

adaptation	recovered (%)		price without error	
	trajectory	deployed	trajectory	deployed
rating error (GAUSSIAN)				
observed-max incumbent	+19* [10, 27]	−10* [−15, −5]	+0.1	+0.2
GP given the true noise variance	−3 [−16, 8]	−3 [−11, 5]	+0.4	−0.0
first ten proposals rated twice	−88* [−113, −67]	−14* [−21, −8]	+5.1	+1.6
+ observed-max incumbent (LogEI)	−93* [−127, −58]	−44* [−57, −32]	+4.7	+1.1
last six trials re-rate the top three	−1* [−2, −1]	+4 [0.04, 9]	+0.1	+0.8
knowledge gradient	−50* [−83, −21]	−39* [−61, −18]	+4.5	+5.3
re-rate the incumbent every second trial	−114* [−155, −76]	−12 [−31, 5]	+7.1	+6.0
unnoticed SLIP				
noisy-input GP	−28* [−43, −13]	−19* [−31, −9]	−0.1	−1.5
last six trials re-rate the top three	0 [−0.3, 0.2]	−6 [−15, 1]	+0.1	+0.8
MISCLICK				
Student- t surrogate, Matérn kernel	−2 [−38, 31]	−13 [−50, 16]	+0.2	+1.5
Student- t likelihood, standard RBF kernel	−58 [−123, −5]	−60 [−114, −10]	+3.4	+5.5
changing what the rater is asked or when				
the rater reports their precision	+16 [−4, 36]	+15* [6, 23]	+0.2	−0.1
each proposal judged beside the best	−18* [−25, −12]	−27* [−36, −19]	+0.0	+0.0
every fifth trial rates an anchor	−6 [−23, 14]	+1 [−8, 10]	+0.8	+0.7
the first five proposals rated late	−13 [−44, 14]	+1 [−11, 11]	+1.6	+0.2
a ship-rule acquisition	−287* [−366, −219]	−107* [−159, −64]	+21.8	+24.4
fitted oracles, rating error				
first ten proposals rated twice	−66 [−78, −49]	−2 [−21, 6]	−	−

clean runs. Against the standard process the conclusions stand, with one addition. The incumbent recovers 19% [10, 27] of the trajectory loss at a price of 0.1 points without error, most of it from the first rating (+10% to +67% across magnitudes), and cuts the extra trials needed, over that arm’s seven acquisitions, to match a clean standard ten-trial study at 1σ from 8 (21% never) to 6 (13%), and at 5σ from 36 (48%) to 21 (39%). The design it deploys is worse, though, by 10% of the cost [5, 15], mostly at the late onset, for a reason we have not isolated. Giving the GP the true noise variance recovers nothing on either response (−3%).

Rating early designs twice makes things worse. The onset effect says the first trials are where error is expensive, so replicating them is the natural fix. It is not one: rating each of the first ten proposals twice recovers −88% of the trajectory loss and −14% of the deployed design’s (Table 34), and it needs more extra trials than the standard process on the same arm to match a clean standard ten-trial study, a median of 8 at 1σ against 6. In a fifty-trial study the ten designs it gives up cost more than averaging two ratings, which only divides the error variance by two, buys. The GP already pools neighbouring observations. Adding the observed-max incumbent of Appendix D.2 does not rescue it (−44% on the deployed design), and on the three fitted-oracle datasets the same replication is indistinguishable from zero on the deployed design (−2%, interval [−21, +6]), though it still costs the trajectory (−66% [−78, −49]). Re-rating the incumbent every second trial fails for the same reason (−114% [−155, −76] on the trajectory). qKG helps only at 5σ from the first rating (+21% on the trajectory) and does not help elsewhere, for a reason we have not tested (Appendix B.8).

Re-evaluating before deploying helps, a little. Spending the last six trials re-rating the three designs with the best mean rating, and deploying by mean, costs almost nothing in the trajectory (−1%) and recovers 4% of the deployed design’s cost, [0.04, 9], reaching 18% [5, 29] when large error arrives late. Under an unnoticed slip it recovers nothing (−6%, [−15, 1]): the re-ratings land on displaced points too, so averaging them does not correct the record.

A noisy-input GP helps small early slips and hurts large late ones. Inflating each point’s observation variance by the squared gradient of the posterior mean times the slip variance (McHutchon & Rasmussen, 2011) recovers 76% [27, 245] of the cost of a 1% slip from the first trial and 17% of a 5% slip, but at the late onset it recovers −129% and −205% of the cost of 15% and 40% slips. A likely reason, which we have not tested, is that by then the search sits near a steep peak whose flanks have the largest gradients, so the observations that locate the peak are discounted most. Pooled it recovers −28% of the trajectory loss.

A heavy-tailed likelihood does not absorb misclicks. A misclick records the rating of a random design against the proposed one, an outlier rather than noise, so a Student- t likelihood should discount it. A variational Student- t GP (Jylänki et al., 2011; Martinez-Cantin et al., 2018) in place of the standard one recovers −2% [−38, 31] of the trajectory loss and −13% [−50, 16] of the deployed design’s, at a price of 0.2 and 1.5 points without error. That GP also swaps the RBF kernel for a Matérn-5/2 and marginal likelihood for a variational bound. Rerun with the standard RBF kernel, so that only the likelihood and its variational fit differ, it does worse: −58% [−123, −5] and −60% [−114, −10], at a price of 3.4 and 5.5 points without error (Table 34). The kernel change did not mask a benefit of the likelihood. Where misclicks cost most, on 40% of trials from the first, the Matérn version recovers −1% [−10, 6] of the trajectory loss and the RBF version +5% [−8, 16], and the median run of either still needs 8 extra trials to match a clean standard ten-trial study, as the standard process does on this arm (24% and 21% never, against 28%). Misclicks cost little in the first place, so there is little for any model to recover.

E.6 FOUR REMEDIES THAT COST NO RATING TRIALS

Every remedy in Appendix E.4 changes the loop. These four change what the study deploys, how it reads the ratings it already has, or what it asks the rater, and three of them are scored by re-reading runs that already exist (`scripts/replay_hitl_remedies.py`), so they cost nothing to evaluate as well as nothing to run. The estimand is the one of Appendix E.5: a share of the standard process’s cost of error on the design the study deploys, as a ratio of landscape means with a landscape bootstrap.

A shortlist instead of a design. Table 35 gives the curve. The gain is entirely the width of the deliverable: at $m = 1$ the same rule recovers nothing on the main sweep. And the score is the regret

Table 35: What a wider deliverable and a rank-based reading recover, on the design the study deploys, as a share of the standard process’s cost of error. The first row ships one design by the posterior mean less one latent SD, for comparison with the wider deliverables below it; the last row applies the same rule on ranks. Twenty landscapes, LogEI and qNEI. The main-sweep column pools four error processes, four magnitudes and both onsets. Gross faults are 15% of trials spiking at 20 landscape SDs, at 0.25σ from the first rating. The capped scale is cut at the landscape’s 0.9 quantile, pooling 0.25σ and 1σ from the first rating.

deliverable	main sweep	gross faults	capped scale
ship 1, posterior mean less one SD	0% [−8, 8]	20% [10, 29]	30% [23, 37]
ship 2	16% [9, 21]	39% [27, 50]	41% [34, 48]
ship 3	24% [18, 30]	47% [35, 58]	46% [40, 54]
ship 5	34% [28, 40]	56% [44, 66]	54% [47, 62]
ship 1, chosen on ranks	−1% [−11, 8]	51% [36, 63]	36% [29, 43]

of the *best* of the m , which a later decision would have to find. It is therefore an upper bound on what shipping a shortlist delivers, to be read as the size of the prize for any downstream discrimination that is cheaper than a human rating. Where no such decision exists the number is unavailable, and a study should say so.

Choosing on ranks. Refitting the surrogate on the ratings’ normal scores instead of their values makes the choice invariant to any strictly monotone distortion of the scale and robust to a single wild rating. It costs 1% on the main sweep, where discarding the magnitudes throws information away, and recovers 51% [36, 63] under gross faults and 36% [29, 43] against a saturating cap on seeds 7–11, and 49% [34, 63] and 38% [31, 45] on seeds 12–16, which were run for this check after the rule was chosen (Appendix E.3). At that fault rate it does at least as well as the relevance-pursuit likelihood (38% [24, 51]) of Appendix E.4 while assuming nothing about the fault. The contrast is the finding: a rank-based reading is not a general improvement, it is an outlier remedy that happens to be free.

Asking the rater how sure they were. A self-reported precision, used as a per-trial observation variance, recovers 15% [6, 23] of the deployed loss on seeds 7–11 at a price of -0.001 (Table 34), and 11% [−2, 23] on seeds 12–16, run for this check after the arm was built. It is worth setting beside the known-noise arm, which is handed the *true* noise variance and recovers nothing: a constant is what the GP already learns, and a per-trial report says which trials were hard. How much it says is the whole question. The first model took the report to be the trial’s squared error seen through a log-normal multiplier of SD 0.5, which gives the rater a rank correlation of about 0.96 with their own realised error, close to clairvoyance. Three further arms name that correlation instead, through a Gaussian copula that keeps the error’s marginal, and run in the two cells where the first model’s recovery is well above zero (0.25σ and 1σ from the first rating, seeds 7–11). Pooled over those cells the first model recovers 31% of the deployed loss; at a correlation of 0.6 the report recovers 12% [−1, 23], at 0.3 it recovers -10% [−25, +4], and an uninformative report, at 0, -16% [−37, +2], presumably because the GP then trusts per-trial variances that are wrong. A study should ask for confidence only if it knows its raters’ confidence tracks their errors.

Screening the instrument. A capped scale costs 185% more than the same noise on a scale that does not saturate, and how often the cap binds is fixed by its quantile. What varies, and what a pilot can measure, is the ceiling gap, how far the reachable optimum sits above the ceiling. Across the twenty landscapes that gap predicts the extra cost at Spearman 0.65, and the extra cost doubles from 0.153 in the half of the landscapes with the smaller gap to 0.316 in the half with the larger gap. Re-anchoring the cap to the best design rated so far halves the median damage, from 0.219 to 0.104. An instrument can therefore be rejected before a participant is spent on it.

What did not work. Six further ideas were built and measured, and they show where the cost lives. Allocating the final sitting’s looks where the ranking is in doubt, by the standard best-arm rule (Audibert et al., 2010; Kim & Nelson, 2006), is never ahead of one look each (+9.0% against +9.7% at $k = 8$): the gain is having looks, not where they go. Sizing a study from the pilot’s frag does

no better than sizing it at random, although any uneven allocation of a fixed budget beats an even one. Interleaving fixed anchor designs to identify the rater’s drift does not help under drift (-6% $[-15, +1]$; $+1\%$ under GAUSSIAN, Table 34) and costs trials. Rating the first five proposals late instead of early moves nothing ($+1\%$), which rules out one explanation of the onset effect. Judging each proposal beside the incumbent, so the error shared by the pair cancels, *loses* 27% : it pays $\sqrt{2}$ more fresh noise to remove faults the loop already absorbs, since it standardises its targets and an argmax is shift-invariant. An acquisition aimed at the ship rule rather than the posterior maximum loses 107% , but its price without error is $+0.244$, the largest in this paper, so it searches badly and its own idea is untested.

E.7 COMPARISONS INSTEAD OF RATINGS: WHAT THE INVARIANCE DOES AND DOES NOT BUY

Preference elicitation is the usual response to an unreliable rater (Chu & Ghahramani, 2005; González et al., 2017), on the argument that a comparison is invariant to how the rater uses the scale. The argument is exact and its reach is narrow, and both halves are measurable here. A fault is *shared* when it acts on every design of one sitting through the same monotone map g ; then for strictly increasing g , $g(f(a)) > g(f(b)) \Leftrightarrow f(a) > f(b)$, and the comparison is exactly what it would have been with no fault. A fault drawn afresh for each judgement cancels nothing, and a comparison inherits the noise of both designs.

We ran two loops at equal human cost on the twenty landscapes, three seeds, $T = 20$ judgements: the standard rating loop, and a comparison loop with a Laplace-approximate PAIRWISEGP (Balandat et al., 2020) on which each trial proposes one design and asks for a comparison with the incumbent, shipping the posterior-mean maximiser. Ten of the 960 fits fell back to prior hyperparameters and are counted, not dropped. Each loop is scored against its own identically seeded clean twin.

The invariance under a shared strictly monotone fault is not a finding but a consequence of the construction above, and the run confirms the implementation rather than the claim: under DRIFT the rating loop’s shipped design differs from its clean twin in 41 of 60 runs, at an excess of 0.040 of the achievable improvement, and the comparison loop’s differs in *none*. What the run does test is the reach of the invariance, and there it is narrow. Under a saturating scale, a map that is monotone but not strictly so, the comparison buys nothing (-0.3% $[-30, +27]$, an interval that spans zero): two designs above the cap are indistinguishable, and near the optimum that is exactly the pair the study must tell apart. Under GAUSSIAN noise, drawn afresh per judgement, the comparison recovers -26.5% $[-79, +15]$, a larger excess, though the interval spans zero. The direction is the one the $\sqrt{2}$ argument predicts, and the magnitude is not established. The three shared faults of this arm carry *no* idiosyncratic noise: the DRIFT here is a bare ramp, the OFFSET a bare constant and the cap a bare truncation, whereas the sweep’s processes of the same name add a GAUSSIAN draw on top. A noise-free strictly monotone fault is the case a comparison is immune to, so the 41-against-none result above states the construction rather than measuring a remedy; with the sweep’s noise added the comparison would inherit $\sqrt{2}$ of it. Read this arm as an upper bound. A constant OFFSET present from the first trial, with no noise added, moves neither loop, in no run, also by construction: a rating loop that standardises its outputs and ships an argmax is already shift-equivariant, so that arm tests the rating loop’s invariance rather than the comparison’s, and the cost of BIAS elsewhere in this paper comes from its noise and from its arriving *part-way through*.

So changing the elicitation is not a general remedy. It removes, exactly, that part of the fault which is a strictly monotone reparametrisation of the scale (drift, a rater’s personal offset, a handover between raters) and nothing of the inconsistency, the saturation or the gross faults.

E.8 DETECTING A MID-STUDY CHANGE, AND WHETHER THE DETECTOR MATTERS

A rater who becomes noisier part-way through a study leaves a trace in the loop’s own surrogate. With its hyperparameters and standardisation frozen on the first $n_0 = 15$ trials, the one-step-ahead standardised residual z_t is standard normal while the rater behaves and its variance inflates when the rater does not. The freeze rule runs a one-sided cumulative-sum (CUSUM) test (Page, 1954) on z_t^2 , alarms at the first t with $S_t > h$, and on alarm stops the study and ships the visited design with the highest posterior mean less one latent SD, refitted on the trials before the alarm window. Its thresholds are tuned on seeds 7–11 to a false-alarm budget of at most 10% on runs in which

nothing changes after n_0 , and scored on seeds 12–16. Pooled over processes, magnitudes and onsets it recovers 14.7% [3, 26] of the deployed loss while saving 4.5 of 50 trials, at a 3.3% false-alarm rate. The effect sits where it should: for a 5σ error arriving at trial 21 it recovers 42% [25, 59] (BIAS) to 64% [51, 75] (DRIFT), detects 94–98% of onsets with a median delay of 3–5 trials and saves 21–24 trials; for a 1σ error arriving at trial 21 detection is 48–58% and recovery -9% to $+7\%$, indistinguishable from zero; for an error present from the first rating there is nothing to detect.

A CUSUM is a choice, not a derivation, so we compare it with two alternatives on exactly the same 13,200 residual streams (6,600 tuning, 6,600 held out; 6,400 carrying a real mid-study change). Since the residual is meant to be standard normal while the rater behaves, and a rater who becomes noisier inflates its variance, every detector tests $\mathcal{N}(0, 1)$ against $\mathcal{N}(0, \sigma^2)$, $\sigma > 1$: the CUSUM; the generalised likelihood ratio (GLR) maximised over the unknown change point (Lorden, 1971), whose statistic $\max_{\tau} \frac{n}{2} (\hat{\sigma}_{\tau}^2 - 1 - \log \hat{\sigma}_{\tau}^2)$ is the classical answer to that question and what a CUSUM approximates; and Bayesian online change-point detection (Adams & MacKay, 2007) with a constant hazard and a normal-inverse-gamma model of the post-change variance, alarming when the posterior probability that the current run is short exceeds a threshold. Each detector’s threshold is tuned on the tuning seeds to a common false-alarm budget (an alarm is false when the rater never changed), and scored on the held-out seeds. Two caveats bound what the comparison can show. The standardised residual is not standard normal while the rater behaves: its root mean square runs from 0.59 to 11.4 across landscapes, so the tuned thresholds are large and the CUSUM and the GLR are both dominated by the same largest squared residual, firing together on all but one of the held-out runs. The budget is also common only in the pool: at a nominal 5% the held-out false-alarm rate is 0% on Branin and 29% on Shekel.

The detector is not the bottleneck. At a 10% budget the GLR detects 45.5% of changes and the CUSUM 45.4%; at 5%, 36.8% against 36.9%; at 20%, 56.0% against 56.4%. The CUSUM matches the classical optimum for this alternative to within half a percentage point at every budget, so there is nothing to be gained by replacing it. Bayesian online detection is worse on detection rate at every budget (23.0%, 32.2% and 45.1% at 5, 10 and 20%) but alarms with a median delay of 2 trials against the CUSUM’s 6–9 pooled over magnitudes. Whether that trade pays depends on how fast the freeze rule’s gain decays with delay, which we have not measured. It is a choice of operating point rather than a better method. What limits the freeze rule is the signal, not the statistic computed from it: a rater who becomes somewhat noisier midway leaves little of one.

F THE PREREGISTERED REPLICATION

F.1 HELD-OUT SEEDS: A VOID REPLICATION AND ITS SENSITIVITY ANALYSIS

Everything above was selected and estimated on the same ten seeds. Before running any fresh ones we wrote down four claims, the statistics that would test them, the thresholds each had to clear, and four control conditions that would void the run, then ran ten fresh seeds under GAUSSIAN error only, as preregistered.

Table 36: Preregistered quantities on the screening seeds and on ten fresh ones. Both columns use GAUSSIAN error only and exclude Rosenbrock, so the screening column differs slightly from the pooled twenty-landscape values of Sections 4–6. The replication line reports the *sensitivity* analysis described in the text, not the preregistered run, which is void.

preregistered quantity	screening	fresh seeds
H1 $\beta(\text{frag})$	+0.65	+0.69
H1 $\beta(\log_{10} \sigma_e)$	−0.06	−0.07
H2 onset ratio	5.44	5.01
H3 robust mean rank	4.24	4.11
H3 fragile mean rank	7.26	6.98
H4 dose-response monotone	yes	yes
replicates: H1: yes; H2: yes; H3: yes; H4: yes		

The preregistered run is void, by its own rule. One control failed: Rosenbrock’s headroom on the fresh seeds is 0.083, below the 0.10 the preregistration fixed in advance. The document says that a control failure makes the run void rather than negative, and an exclusion applied after seeing the data cannot undo that. We report it as void.

We also report what it showed, labelled as the post-hoc sensitivity analysis it is. Dropping Rosenbrock from both arms, all four claims clear their thresholds: $\beta(\text{frag})$ moves from +0.65 to +0.69 while $\beta(\log_{10} \sigma_e)$ moves from -0.06 to -0.07 with an interval containing zero. The onset ratio is 5.01 against 5.44. The robust and fragile acquisition families sit at mean ranks 4.11 and 6.98 against 4.24 and 7.26; and the dose-response is monotone in both (Table 36). One deviation and two caveats should be named. The script tested the family contrast over the eight conditions rather than the twenty landscapes as preregistered; run as preregistered it gives $p = 3.6 \times 10^{-5}$ over the twenty landscapes and 7.2×10^{-5} without Rosenbrock. H2 tests the onset ratio of the trajectory loss, which Appendix B.3 shows to be the bound of the clean run’s post-onset improvement, so its replication confirms the bound rather than an effect of the onset. And the null on the error magnitude holds under the preregistered additive specification only: with the magnitude-by-opt_z interaction of Appendix C.2 the magnitude is significant again, so that sub-claim confirms a specification, not a fact about the error.

The headroom control exists to stop a landscape with nothing to lose from contributing a null that looks like evidence of robustness. Five of the sub-claims are positive results, and including a low-headroom landscape makes those harder to obtain, not easier. The two null-type sub-claims, that the magnitude’s interval contains zero and that Kendall’s W is below 0.40, run the other way, and there the sensitivity estimates are within 0.01 of the void run’s (interval $[-0.13, 0.09]$ against $[-0.13, 0.08]$; W 0.272 against 0.276). That is a reason to read the sensitivity analysis as informative. It is not a reason to call the run confirmed, and we do not.